AI for Social Good

Bringing AI Research to Deployment

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Today's Lecture

- Overview of AI for Social Good
- Example projects
  - PAWS and NewsPanda: AI for Environmental Conservation
  - Improving Operational Efficiency of Food Rescue Platforms
  - AI for Mental Health and Behavioral Health
- Summary

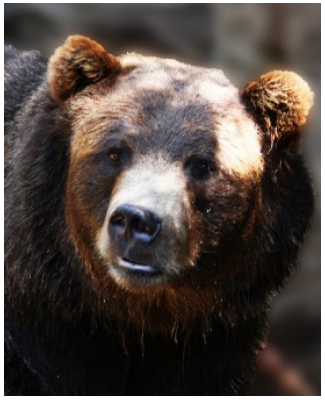
AI for Social Good

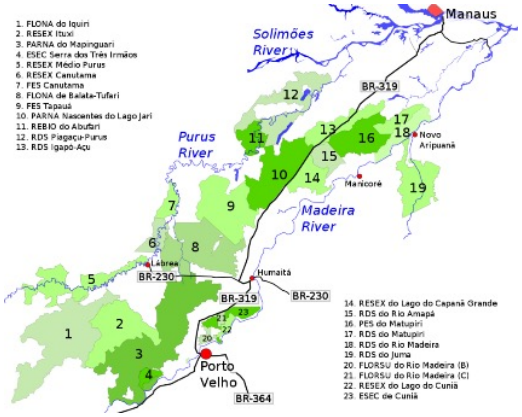
- AI research that can deliver societal benefits now and in the near future



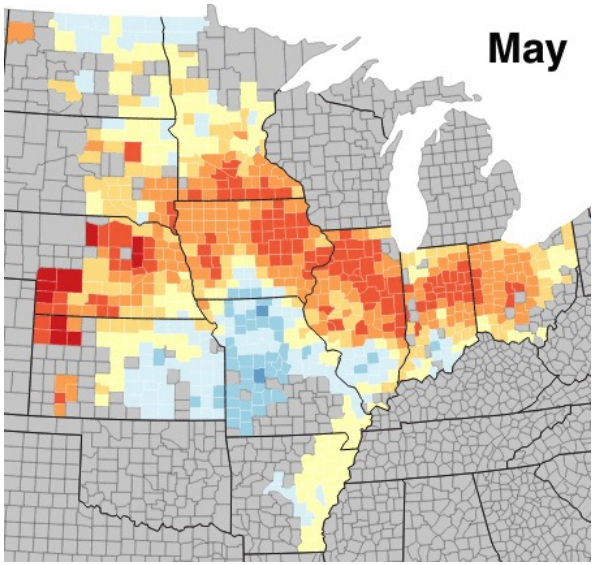
AI for Social Good

Wildlife Corridor Design

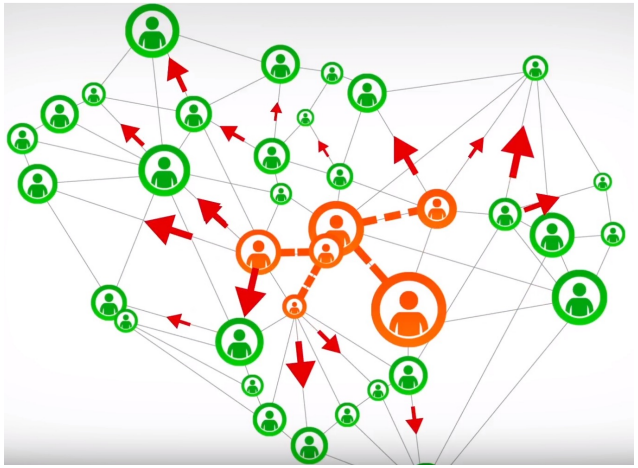




Crop Yield Prediction



Suicide Prevention



Smart Traffic Light Control





https://www.pastemagazine.com/articles/2017/04/a-new-smart-technology-will-help-cities-drasticall.html
http://mashable.com/2015/02/06/hiv-homeless-teens-algorithm/#..k9dRKhxaqm

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Protect Ferries from Attacks



US Coast Guard

Stackelberg game: Leader – Defender, Follower – Attacker

$$\min_{p_1,p_2,\ldots,p_R} \max_{i,\hat{t}} \text{Attacker's Expected Utility}(i,\hat{t})$$

Adversary

Defender		10:00:00 AM Target 1	10:00:01 AM Target 1	…	10:30:00 AM Target 3	…
30%	Purple Route	-5, 5	-4, 4		0, 0	
40%	Orange Route					
20%	Blue Route					
	……					

Deployed by US Coast Guard Since 2013



Fang, F., Jiang, A.X. and Tambe, M., 2013. Protecting moving targets with multiple mobile resources. JAIR, 48, pp.583-634.

How to better protect wildlife using AI?









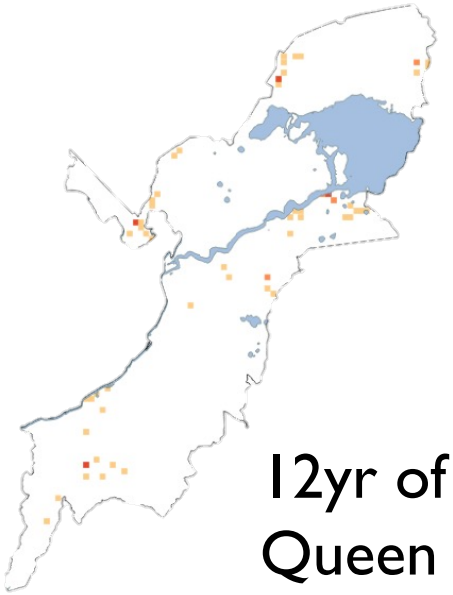


Protect Wildlife from Poaching

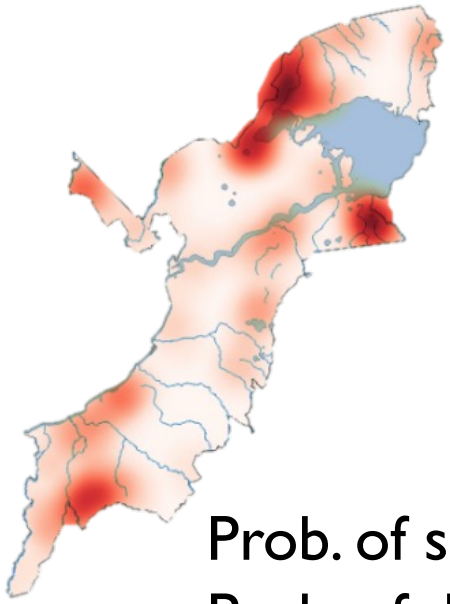
- Leader – Conservation agency
- Follower – Poachers
- New Challenge: Poacher Behavior







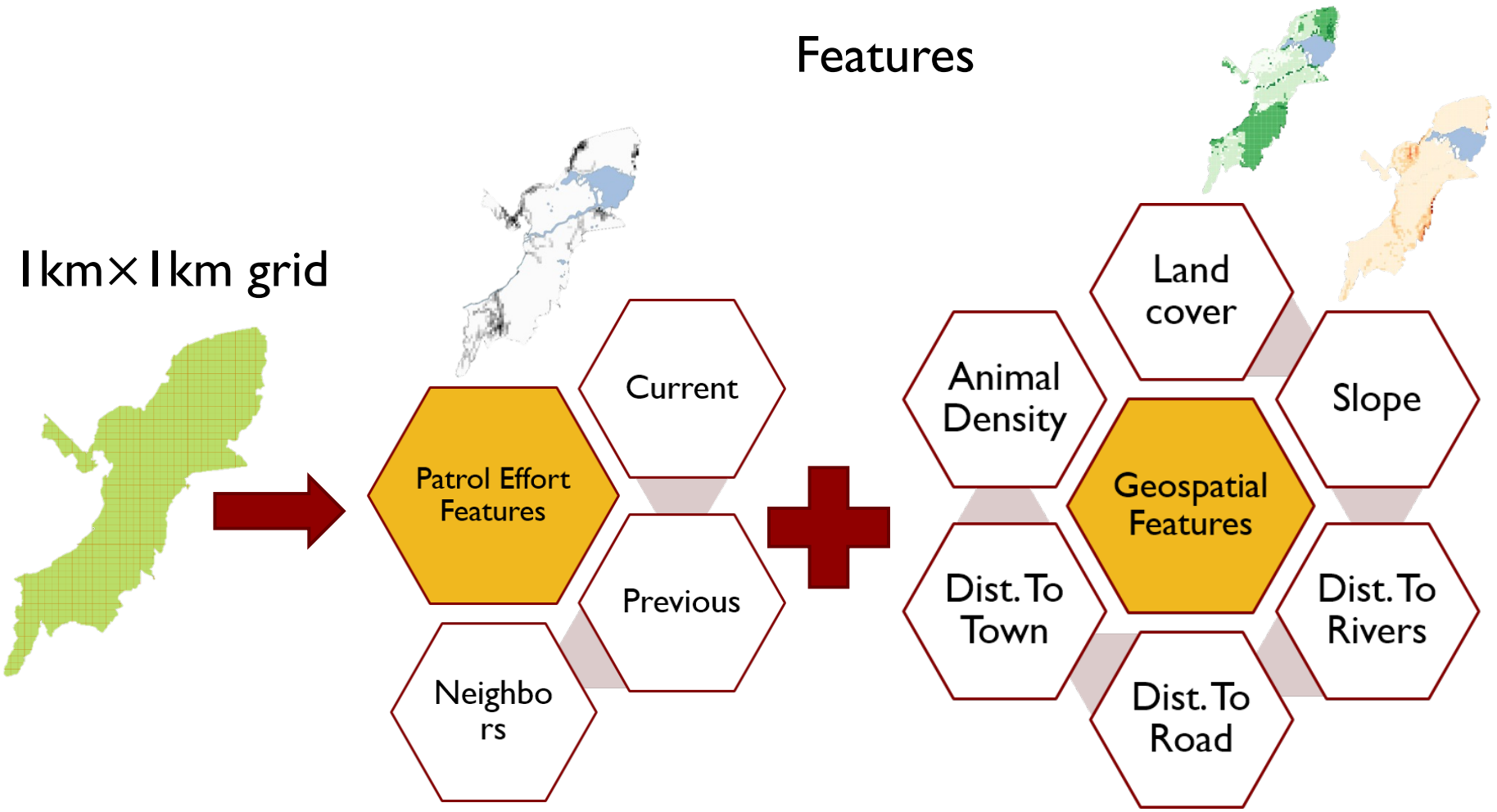
12yr of poaching data from Queen Elizabeth National Park in Uganda



Prob. of snaring or
Prob. of detection

With Uganda Wildlife Authority and Wildlife Conservation Society

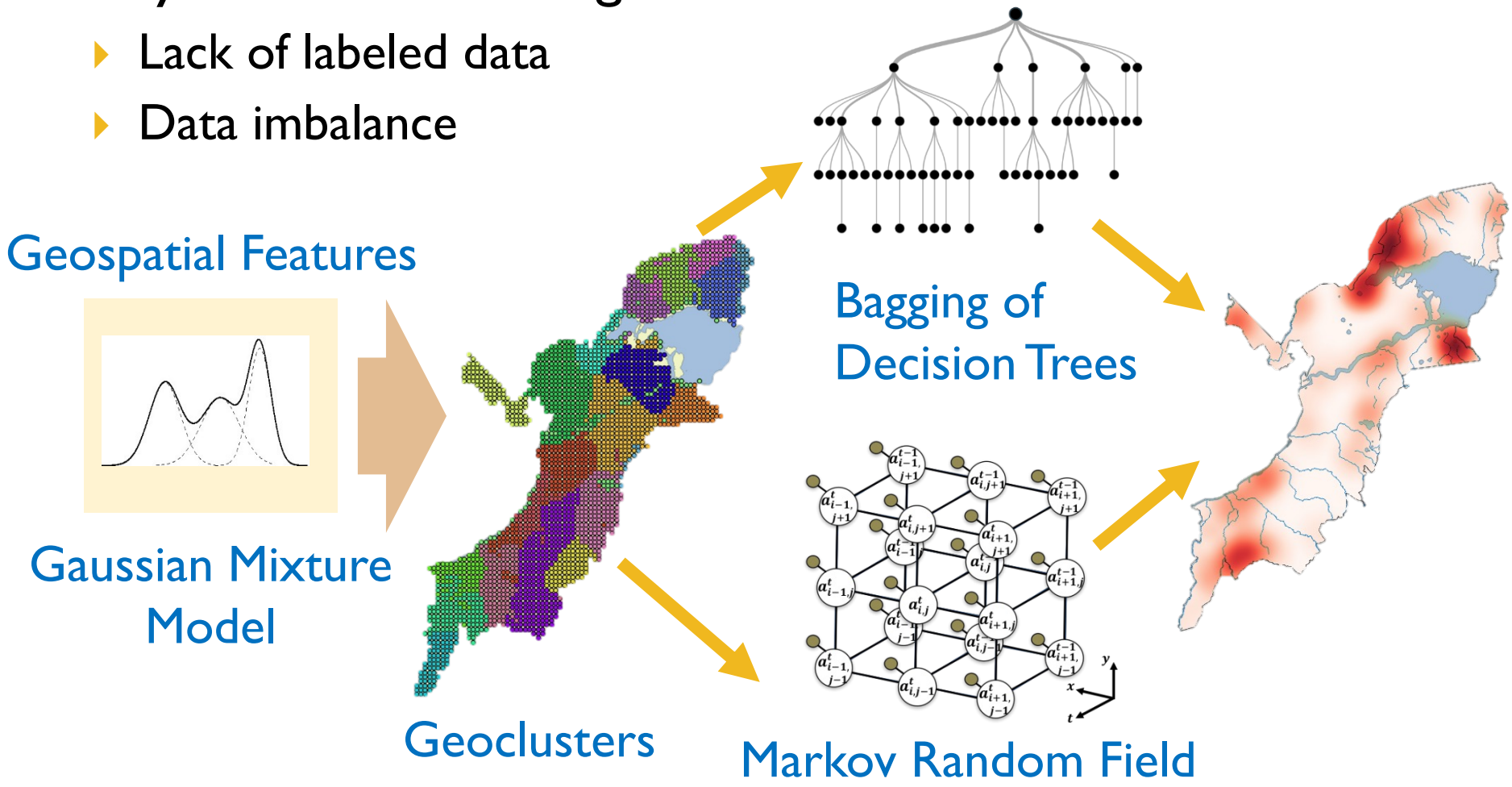
Learn Poacher Behavior from Real-World Data



With Uganda Wildlife Authority and Wildlife Conservation Society

Learn Poacher Behavior from Real-World Data

- Key technical challenges
  - Lack of labeled data
  - Data imbalance

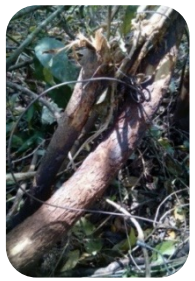


Taking it for a Test Drive: A Hybrid Spatio-temporal Model for Wildlife Poaching Prediction Evaluated through a Controlled Field Test. Shahrzad Gholami, Benjamin Ford, Fei Fang, Andrew Plumptre, Milind Tambe, Margaret Driciru, Fred Wanyama, Aggrey Rwetsiba, Mustapha Nsubaga, Joshua Mabonga. In ECML-PKDD 2017

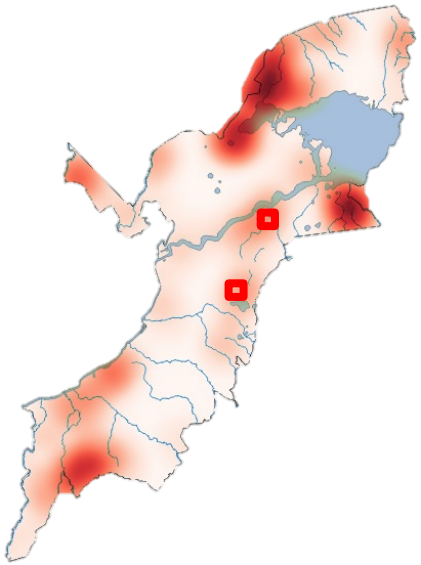
1 Month Field Test in Uganda

- Findings
  - 19 litter, ashes, etc.
  - 1 poached elephant
  - 1 active snare
  - 10 antelope snares
  - 1 roll of elephant snares







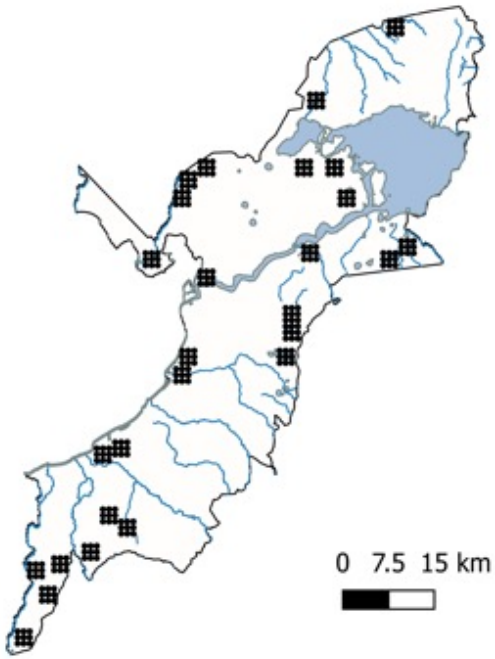


Historical Base Hit Rate	Our Hit Rate
Average: 0.73	3 (>91% months)

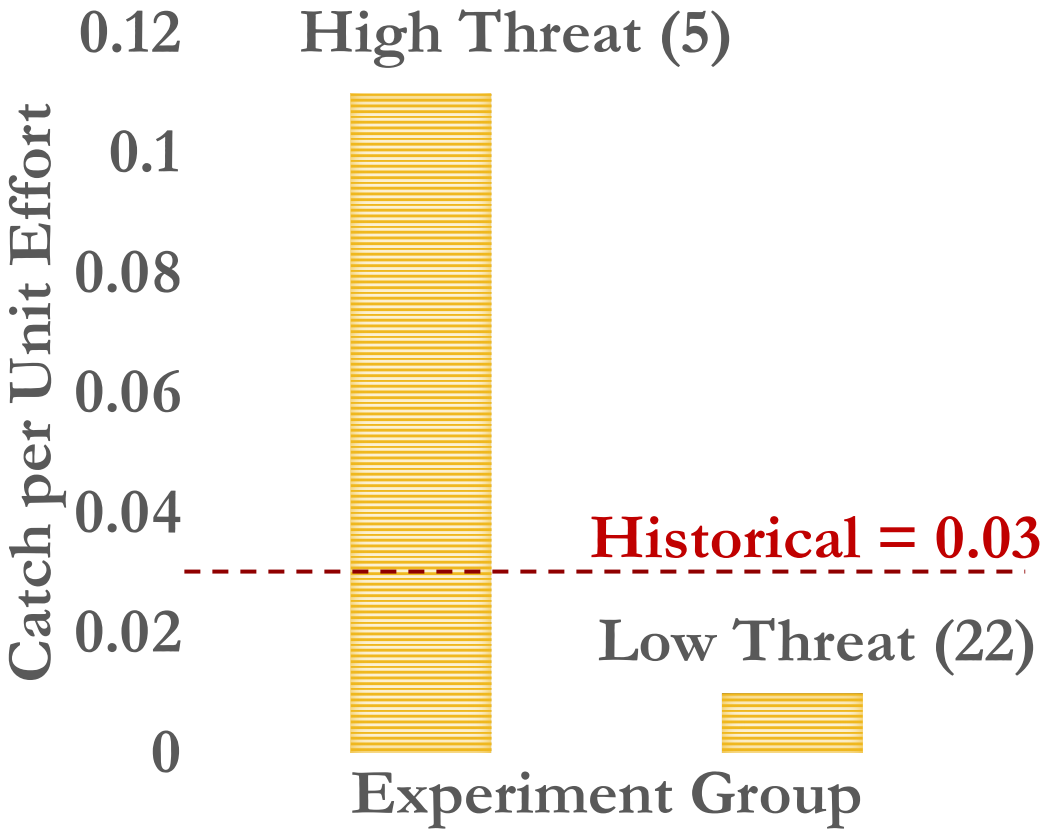


Kar, D., Ford, B., Gholami, S., Fang, F., Plumptre, A., Tambe, M., Driciru, M., Wanyama, F., Rwetsiba, A., Nsubaga, M. and Mabonga, J., 2017. Cloudy with a chance of poaching: Adversary behavior modeling and forecasting with real-world poaching data.

8 Month Field Test in Uganda



- 27 areas
- 452 km patrolled
- Can differentiate H/L threat areas



Taking it for a Test Drive: A Hybrid Spatio-temporal Model for Wildlife Poaching Prediction Evaluated through a Controlled Field Test. Shahrzad Gholami, Benjamin Ford, Fei Fang, Andrew Plumptre, Milind Tambe, Margaret Driciru, Fred Wanyama, Aggrey Rwetsiba, Mustapha Nsubaga, Joshua Mabonga. In ECML-PKDD 2017

Improving the Learning Framework

- Extract geospatial feature from satellite imagery
- Exploit human experts' knowledge

Data from patrols and satellite imagery

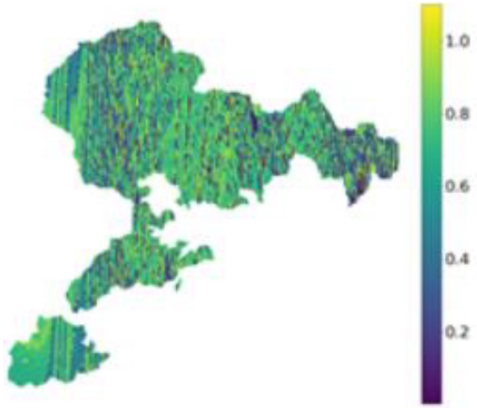


Questionnaire for Rangers

Machine Learning Methods

Ensemble Learning, Decision Trees, Neural Networks, Gaussian Process, Markov Random Field, …

Predicted poaching threat



With the World Wide Fund for Nature and Huang Ni He Forest Bureau

Field Tests in China

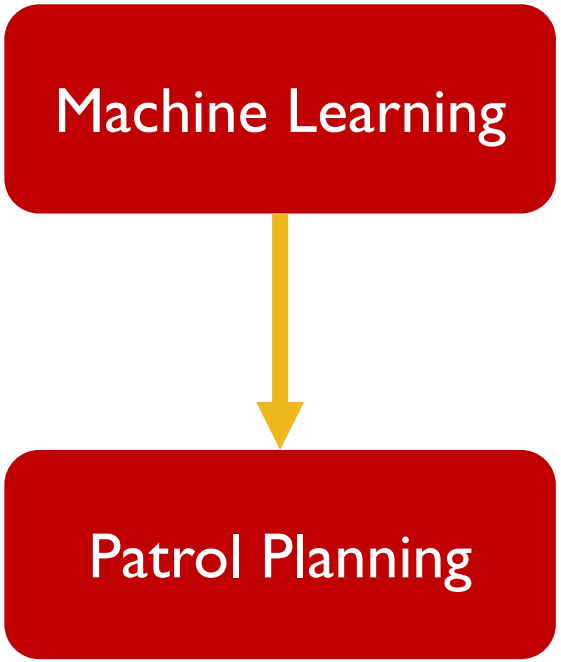
- 2-day field test in 2017: 22 snares
- 2017 winter season: 34 patrol shifts, 7 snares
- 1-week field test in 2019: 42 snares, 3 encounters



Gurumurthy et al. Exploiting Data and Human Knowledge for Predicting Wildlife Poaching. In COMPASS-18; Chen, W., et al. Data-Driven Multimodal Patrol Planning for Anti-poaching. In IAAI-21.

Patrol Planning Based on Learned Poacher Model

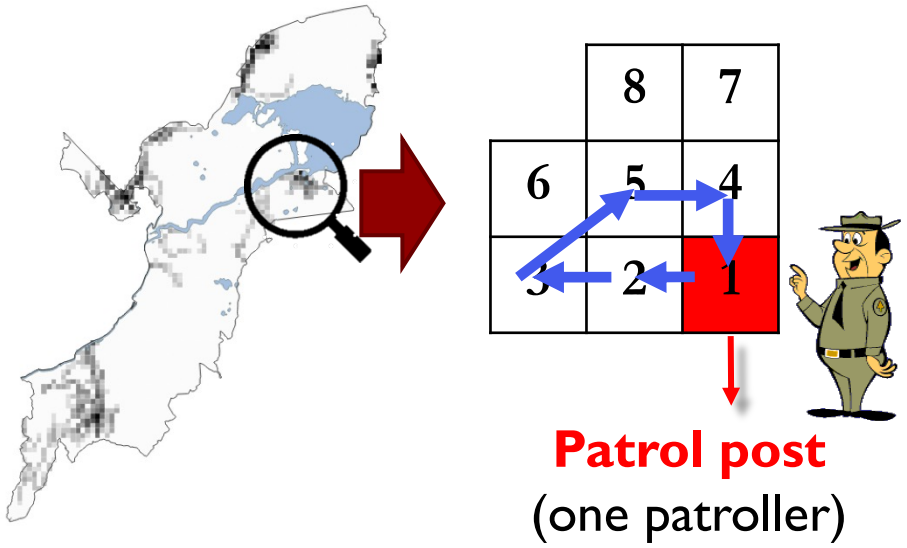




Patrol Planning Based on Learned Poacher Model

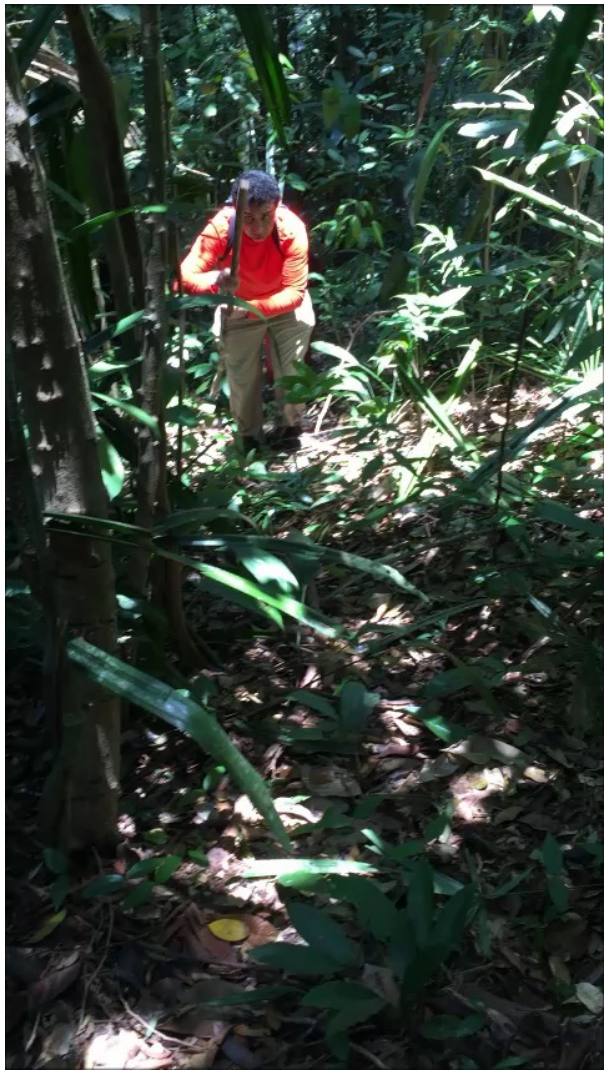


- Maximize detection
- Scheduling constraint



Xu, H., et al. Optimal patrol planning for green security games with black-box attackers. In GameSec 2017

Complex Terrain

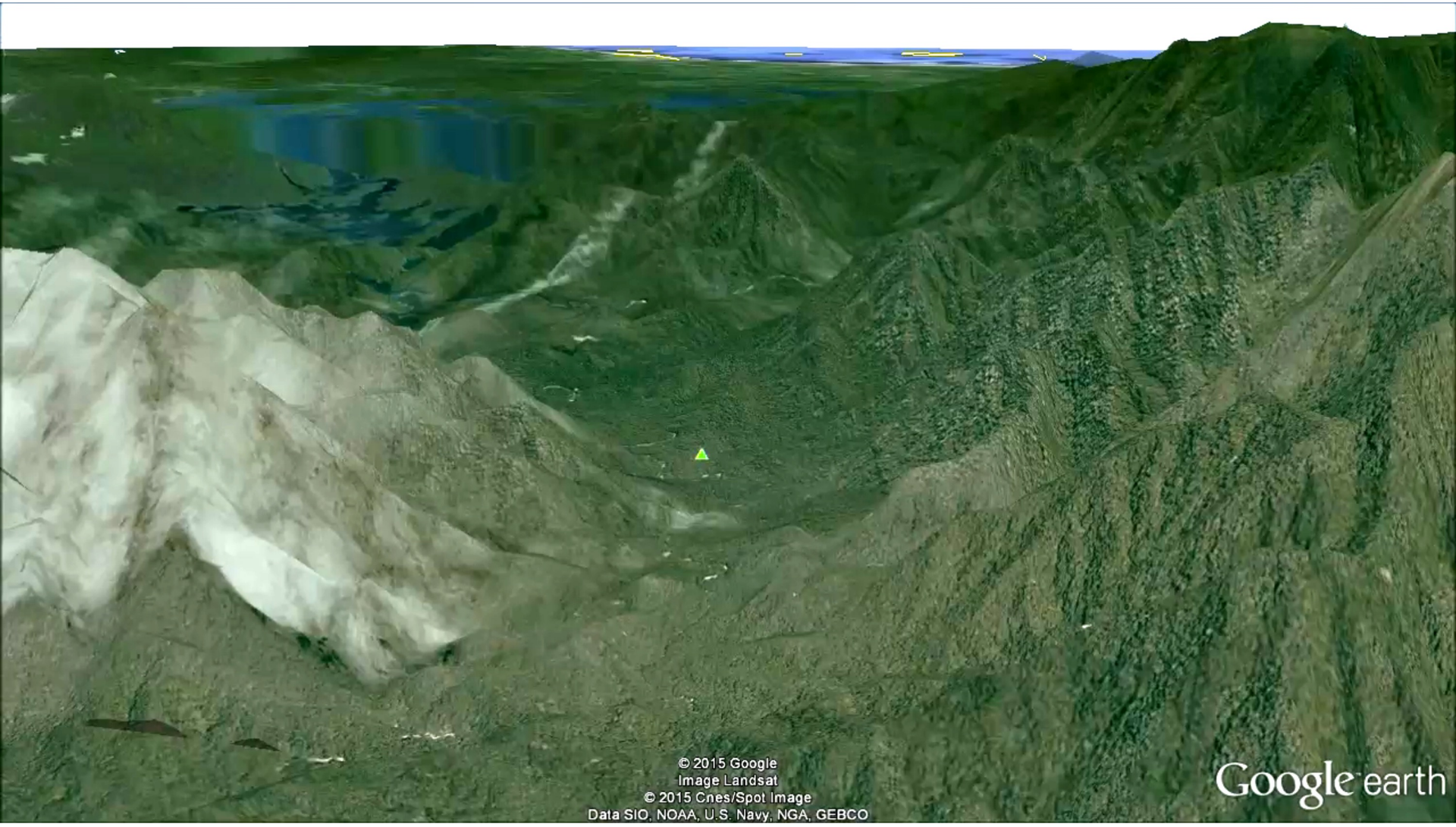






With Panthera and Rimba

Route Planning with Terrain Information



PAWS PATROLS IN THE FIELD

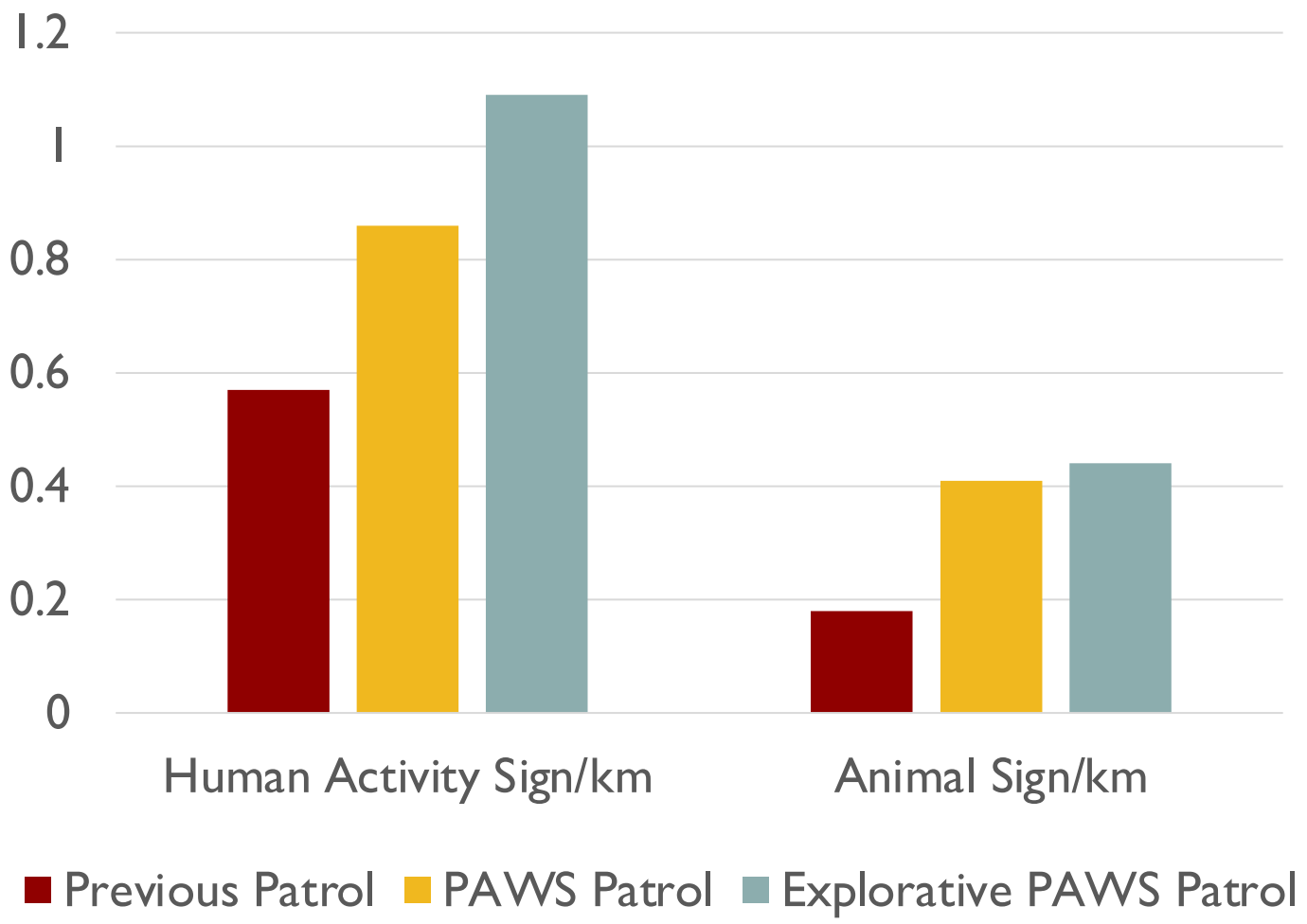
Basic Information of PAWS Patrols	
Average Trip Length	4.67 Days
Average Number of Patrollers	5
Average Patrol Time Per Day	4.48 hours
Average Patrol Distance Per Day	9.29 km



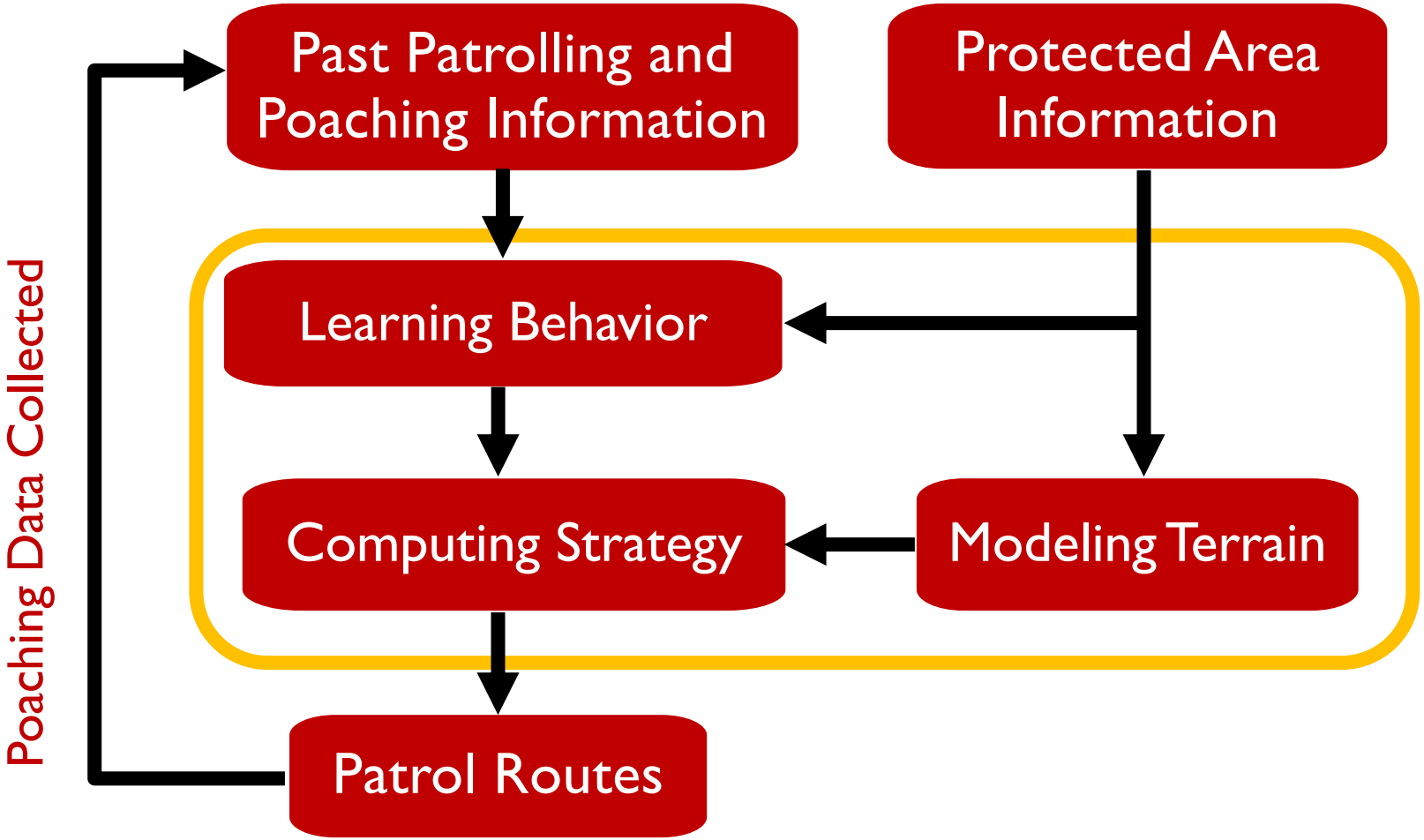




PAWS PATROLS IN THE FIELD



PAWS: Protection Assistant for Wildlife Security



Deploying PAWS: Field Optimization of the Protection Assistant for Wildlife Security
Fei Fang, Thanh H. Nguyen, Rob Pickles, Wai Y. Lam, Gopalasamy R. Clements, Bo An, Amandeep Singh, Milind Tambe, Andrew Lemieux
In IAAI-16: The Twenty-Eighth Annual Conference on Innovative Applications of Artificial Intelligence, February 2016

PAWS: Protection Assistant for Wildlife Security

- Successfully integrated into SMART
- Available to more than 600 sites worldwide
- Already in use in some parks (e.g., Cambodia)















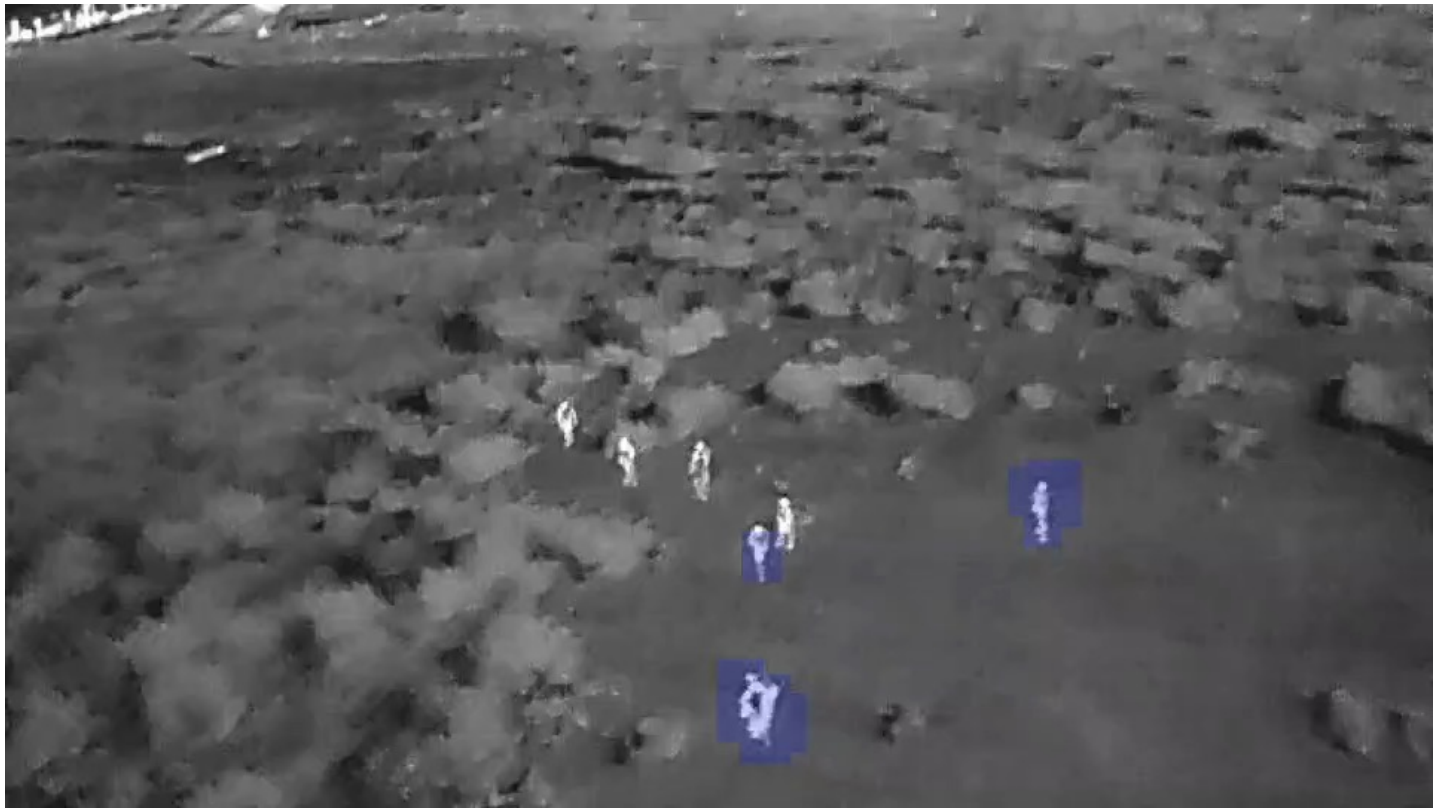






UAV & Human Coordinated Patrols

- Deep learning-based detection
- Game-theoretic models for strategic signaling



SPOT Poachers in Action: Augmenting Conservation Drones with Automatic Detection in Near Real Time. Bondi et al. In IAAI-18

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NewsPanda – Media Monitoring for Conservation

- WWF look through news articles to identify trends, events, or threats related to conservation
  - Identifying 7-10 articles can take 2-3 days to complete

- How do we identify and analyze media articles **automatically** for timely conservation and infrastructure actions?

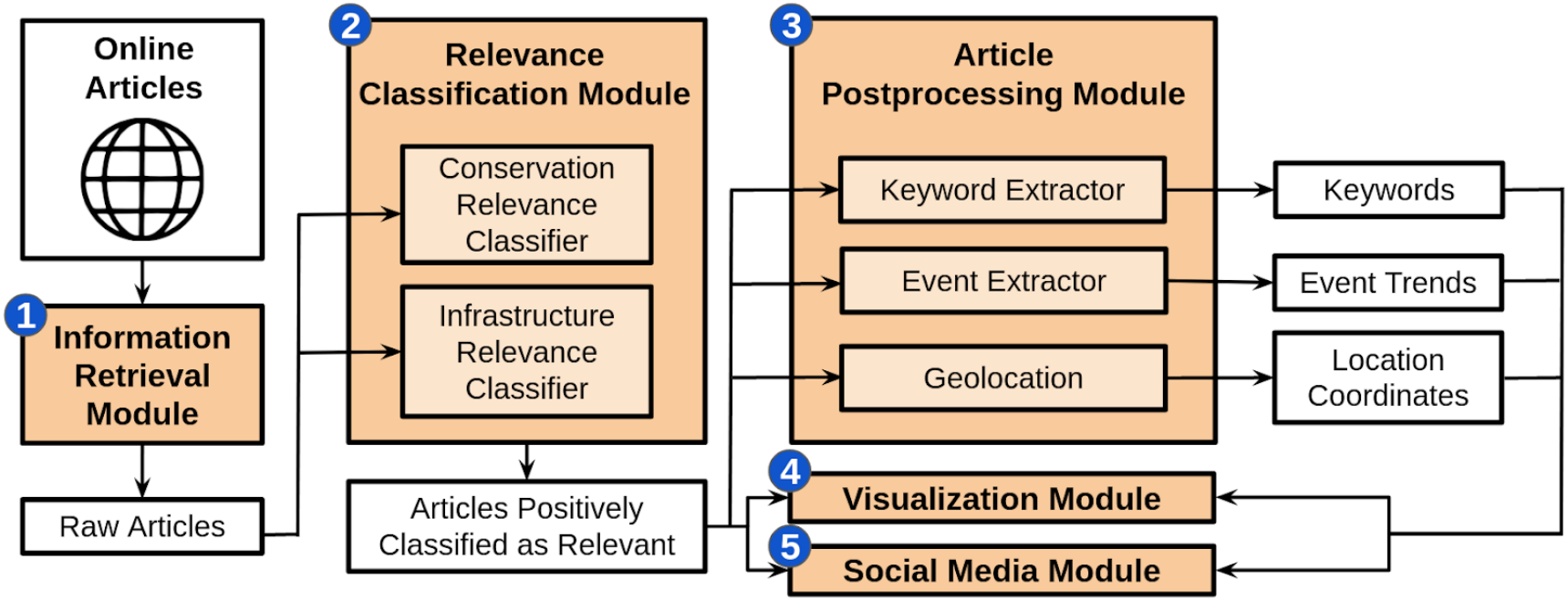


NewsPanda: Media Monitoring for Timely Conservation Action. Sedrick Scott Keh*, Zheyuan Ryan Shi*, David J. Patterson, Nirmal Bhagabati, Karun Dewan, Areendran Gopala, Pablo Izquierdo, Debojyoti Mallick, Ambika Sharma, Pooja Shrestha, Fei Fang. IAAI-23

NewsPanda – Media Monitoring for Conservation

- Fine-tune language model (BERT) for classification
- Active learning + Noisy label correction
- Deployed by WWF in India&Nepal&UK since Feb'22

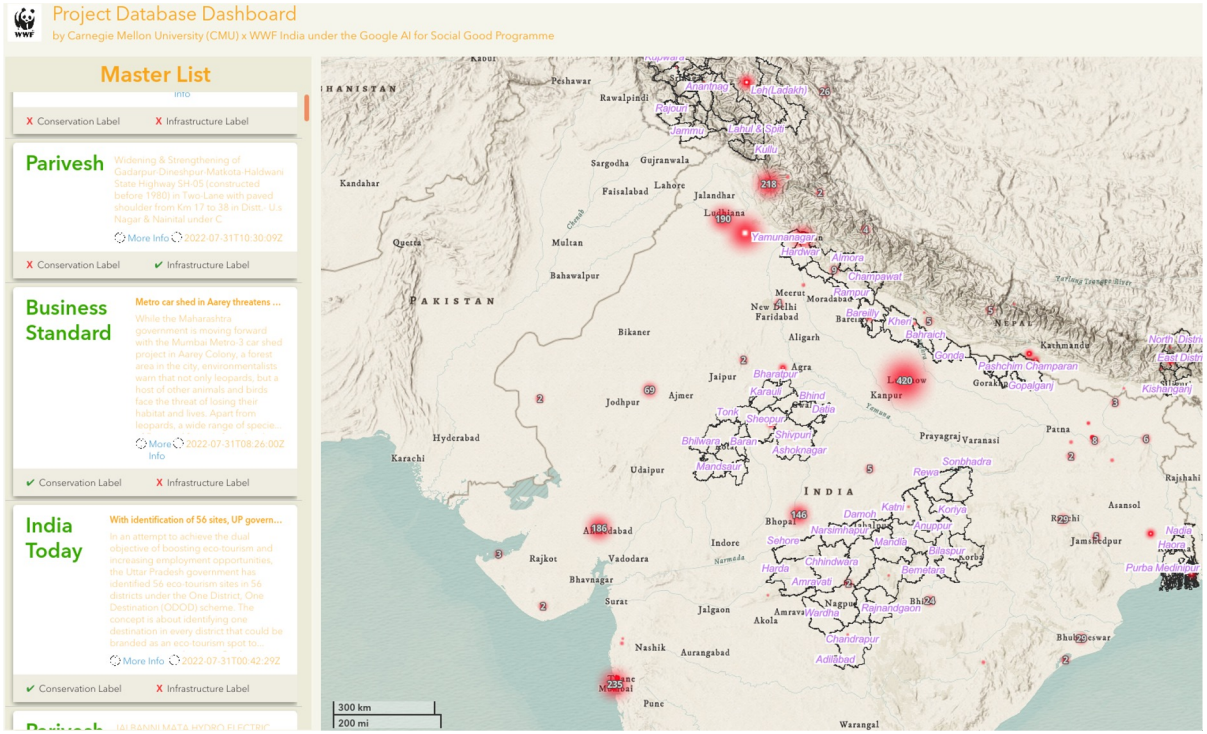
This entire pipeline is ran on a weekly basis.



NewsPanda: Media Monitoring for Timely Conservation Action. Sedrick Scott Keh*, Zheyuan Ryan Shi*, David J. Patterson, Nirmal Bhagabati, Karun Dewan, Areendran Gopala, Pablo Izquierdo, Debojyoti Mallick, Ambika Sharma, Pooja Shrestha, Fei Fang. IAAI-23

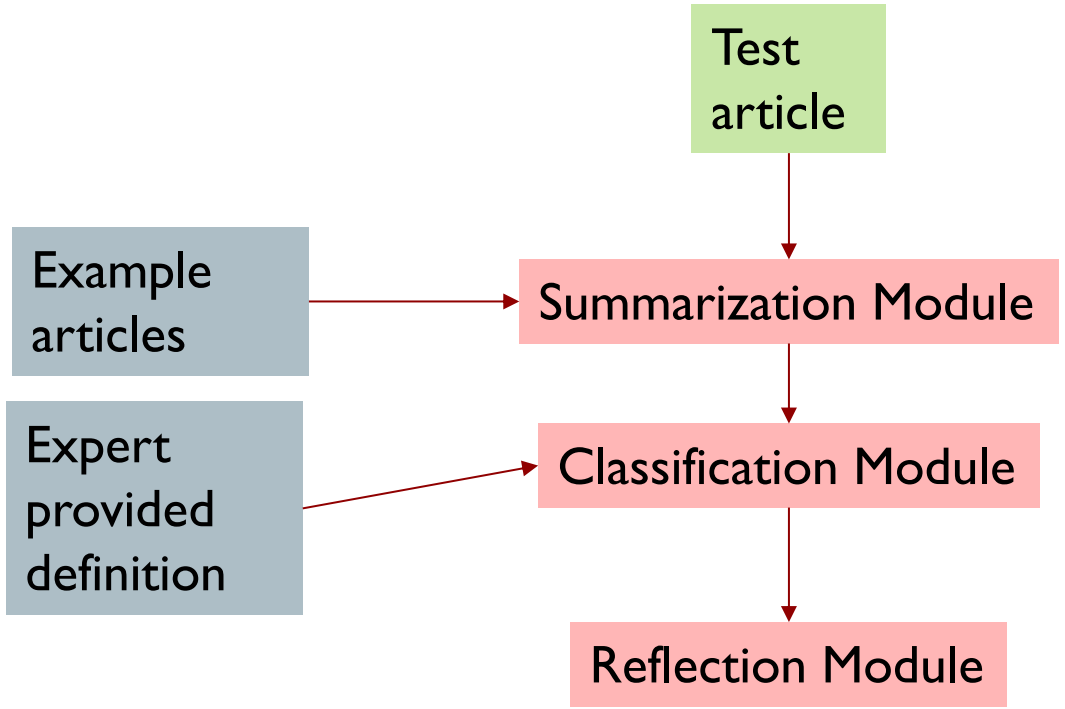
Visualization Module

- The NewsPanda results are integrated into WWF's GIS systems, which is especially useful for the field teams.



How about Conservation Monitoring for Low-Resource Languages?

- Challenge: Extremely limited labeled data
- NewsSerow: Leverage the power of LLMs!



Sameer Jain, Sedrick Scott Keh, Shova Chhetri, Karun Dewan, Pablo Izquierdo, Johanna Prussmann, Pooja Shrestha, Cesar Suarez, Zheyuan Ryan Shi, Lei Li, Fei Fang. Where It Really Matters: Few- Shot Environmental Conservation Media Monitoring for Low-Resource Language. AAAI-24

Deployment

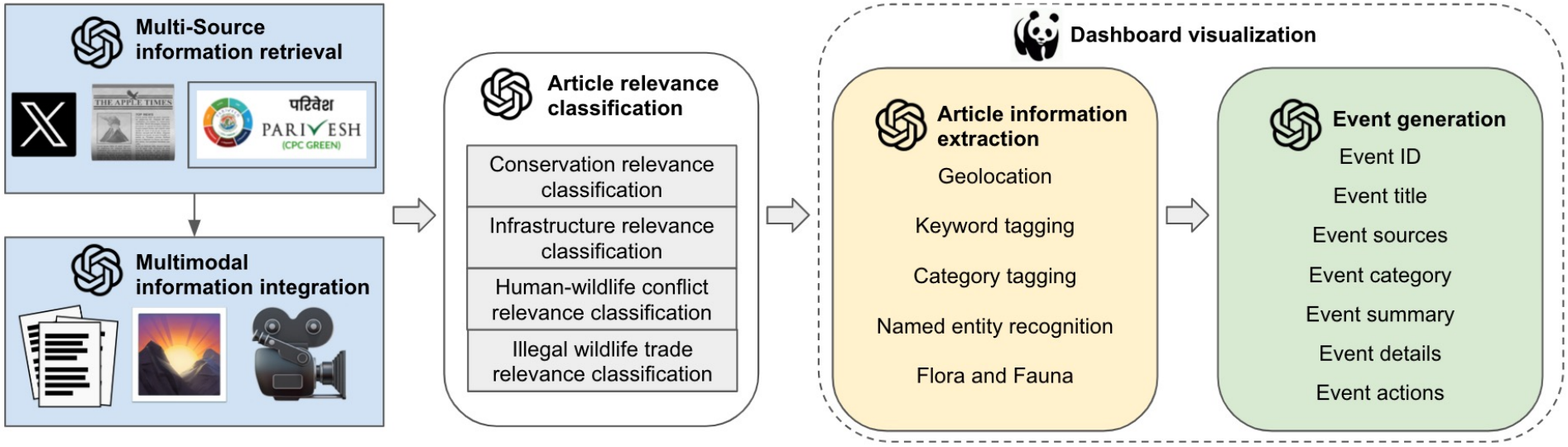
- NewsSerow has been deployed at WWF offices in Nepal and Colombia since Apr 2023 (latest version in use since Aug 2023)
- Results from 8 weeks of deployment:

	Nepal/Nepali				Colombia/Spanish			
Week	**# Ex.**	**P**	**R**	**F1**	**# Ex.**	**P**	**R**	**F1**
1	10	0.5	1.00	0.67	28	0.57	1.00	0.73
2	9	0.00	0.00	0.00	22	1.00	0.70	0.82
3	21	1.00	0.33	0.50	28	0.5	0.8	0.62
4	9	1.00	0.75	0.86	26	0.63	0.71	0.67
5	9	0.50	0.50	0.50	35	0.67	0.80	0.73
6	5	0.50	1.00	0.67	23	0.38	0.50	0.43
7	8	1.00	1.00	1.00				
8	13	1.00	0.38	0.55				
Aggr.	**84**	**0.77**	**0.55**	**0.64**	**162**	**0.61**	**0.73**	**0.67**

Go Beyond News Aricles: Multimodal Social Media Posts



NewsPanda++


Multi-Source information retrieval
THE APPLE TIMES
पटिवेश
PARIVESH
(CPC GREEN)
Multimodal information integration
Article relevance classification
Conservation relevance classification
Infrastructure relevance classification
Human-wildlife conflict relevance classification
Illegal wildlife trade relevance classification
Dashboard visualization
Article information extraction
Geolocation
Keyword tagging
Category tagging
Named entity recognition
Flora and Fauna
Event generation
Event ID
Event title
Event sources
Event category
Event summary
Event details
Event actions

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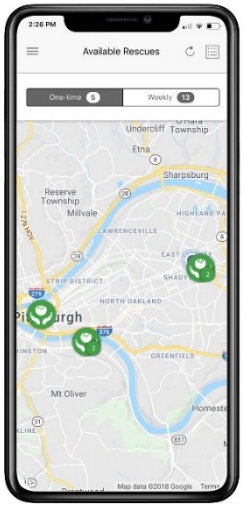
Volunteer-Based Food Rescue Platform

- Food waste and food insecurity coexist
  - Waste up to 40% food globally
  - 1 in 8 people go hungry every day
- Rescue good food!













In collaboration with 412 Food Rescue (412FR)

Volunteer-Based Food Rescue Platform

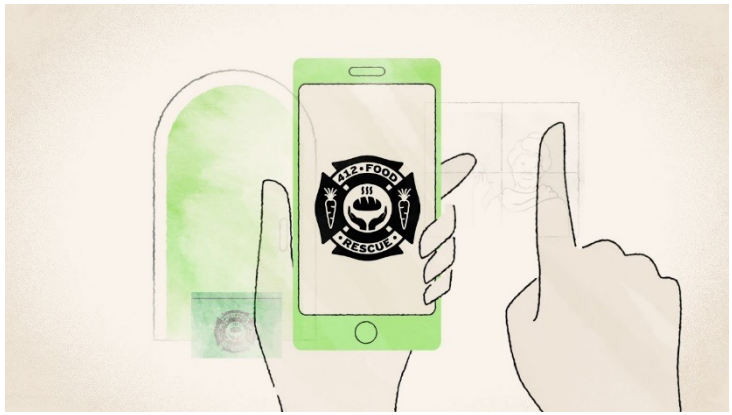
- Our focus: Platform and volunteers
- Key Challenge: Uncertainty about whether a rescue will be claimed and completed

Send notifications to volunteers



- 1-to-many communication
- Volunteer: notification fatigue

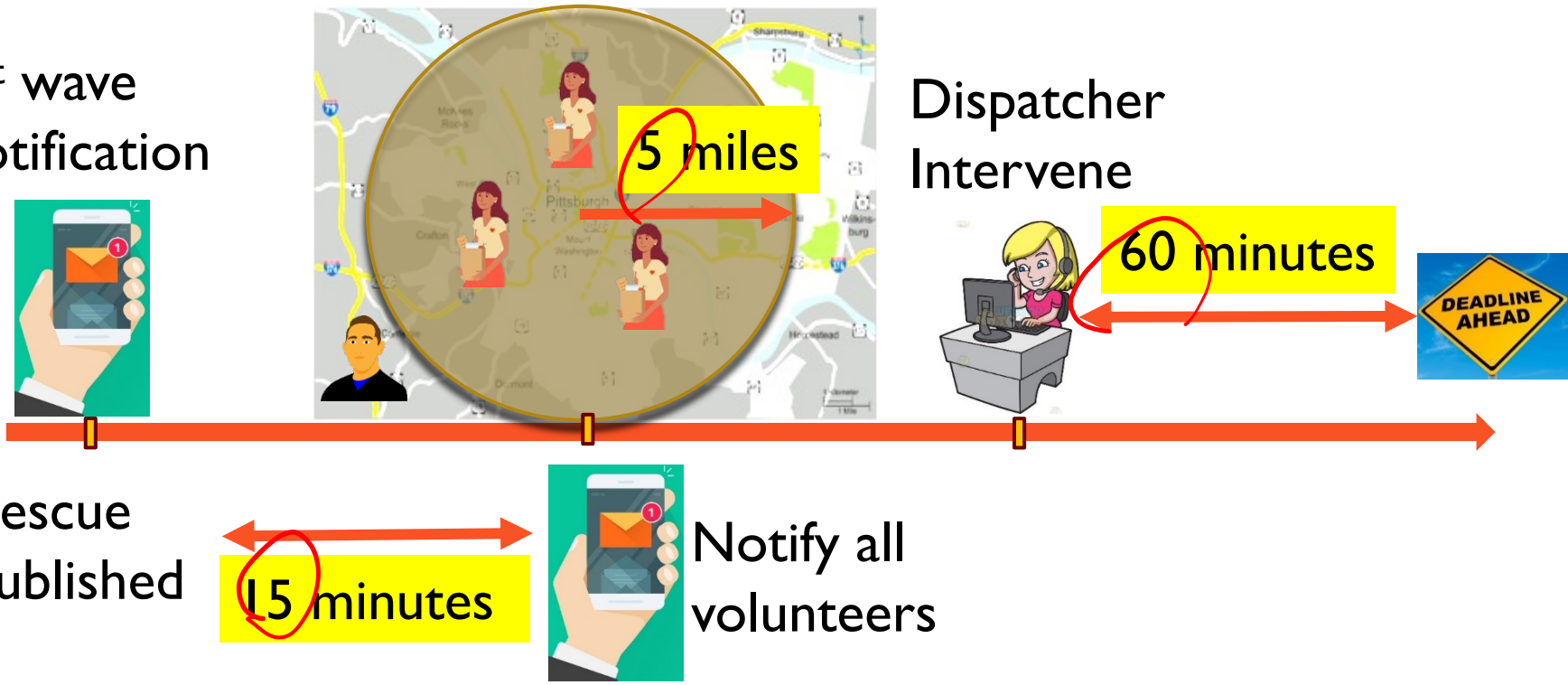
Human dispatcher intervenes



- 1-to-1 communication
- Dispatcher: overstretched

Optimize Intervention and Notification Scheme (INS)

Current Practice (Default INS)



964 volunteers get 1st-wave notification on average

44.6% rescues are claimed by volunteers receiving 1st-wave notification

Improving Efficiency of Volunteer-Based Food Rescue Operations. Zheyuan Ryan Shi*, Yiwen Yuan*, Kimberly Lo, Leah Lizarondo, Fei Fang. In IAAI-20.

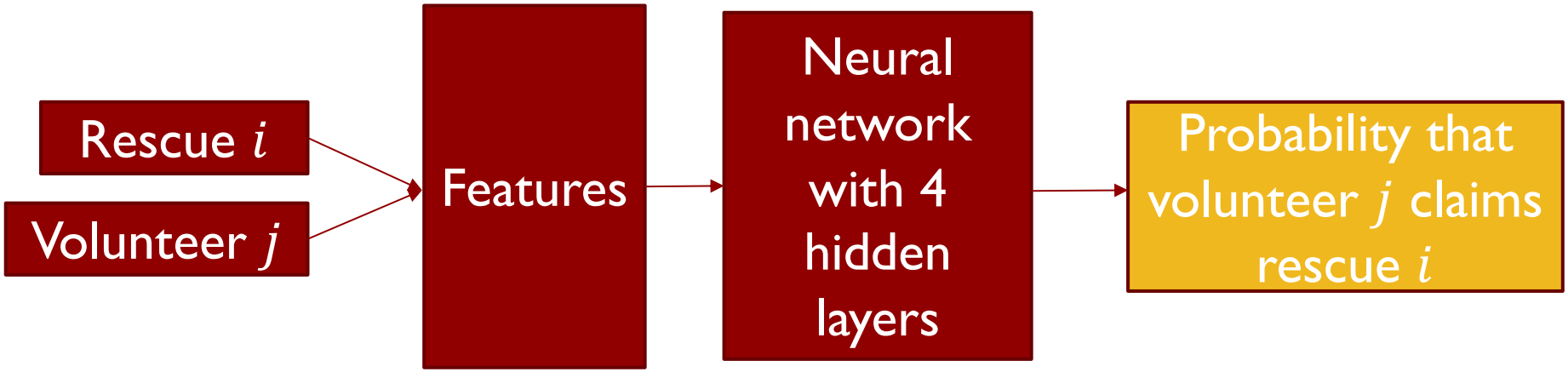
Predictive Model of Rescue-Volunteer Compatibility

Predict whether a rescue will be claimed by a specific volunteer

+: Claimed
-: Not claimed

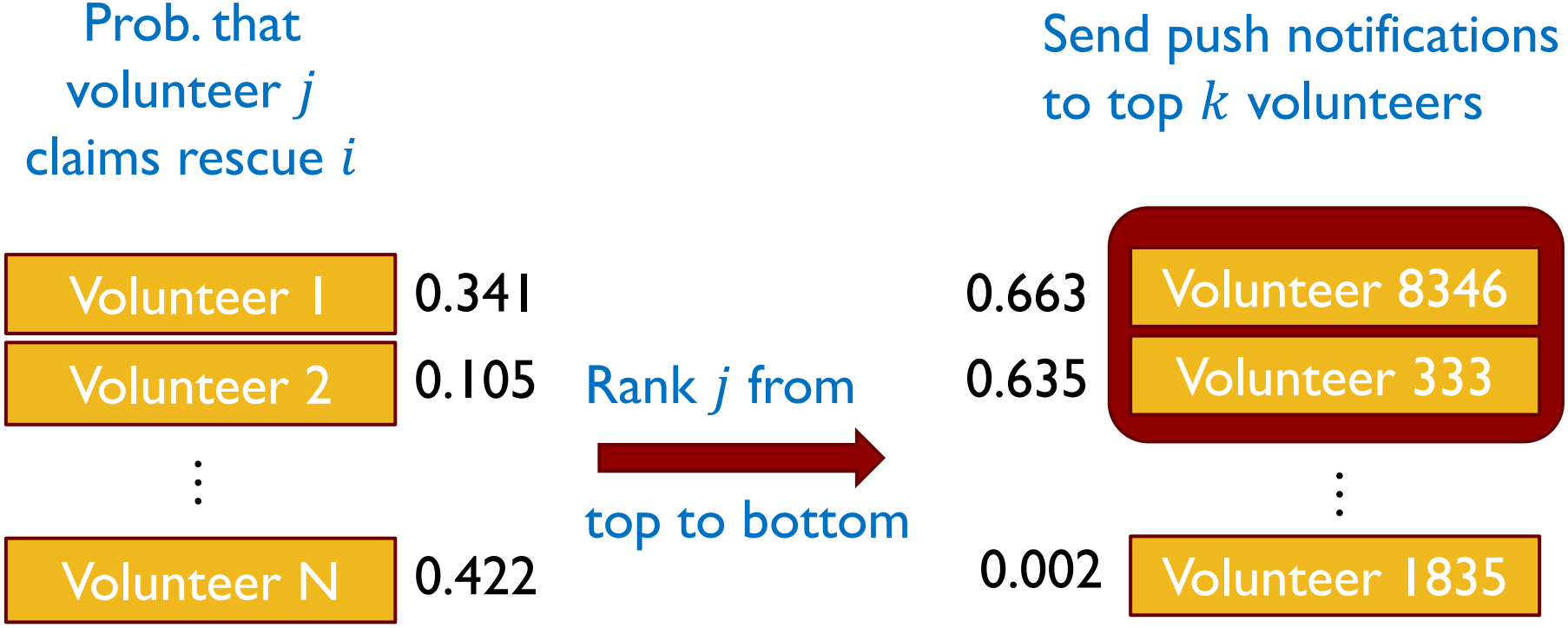
6757 rescues
9212 volunteers

Training data: Mar 2018 to Oct 2019
Test data: Nov 2019 to Mar 2020



A Recommender System for Crowdsourcing Food Rescue Platforms. Zheyuan Ryan Shi, Leah Lizarondo, Fei Fang In WWW-21

Rescue-Specific Notification



A Recommender System for Crowdsourcing Food Rescue Platforms. Zheyuan Ryan Shi, Leah Lizarondo, Fei Fang In WWW-21

Evaluation

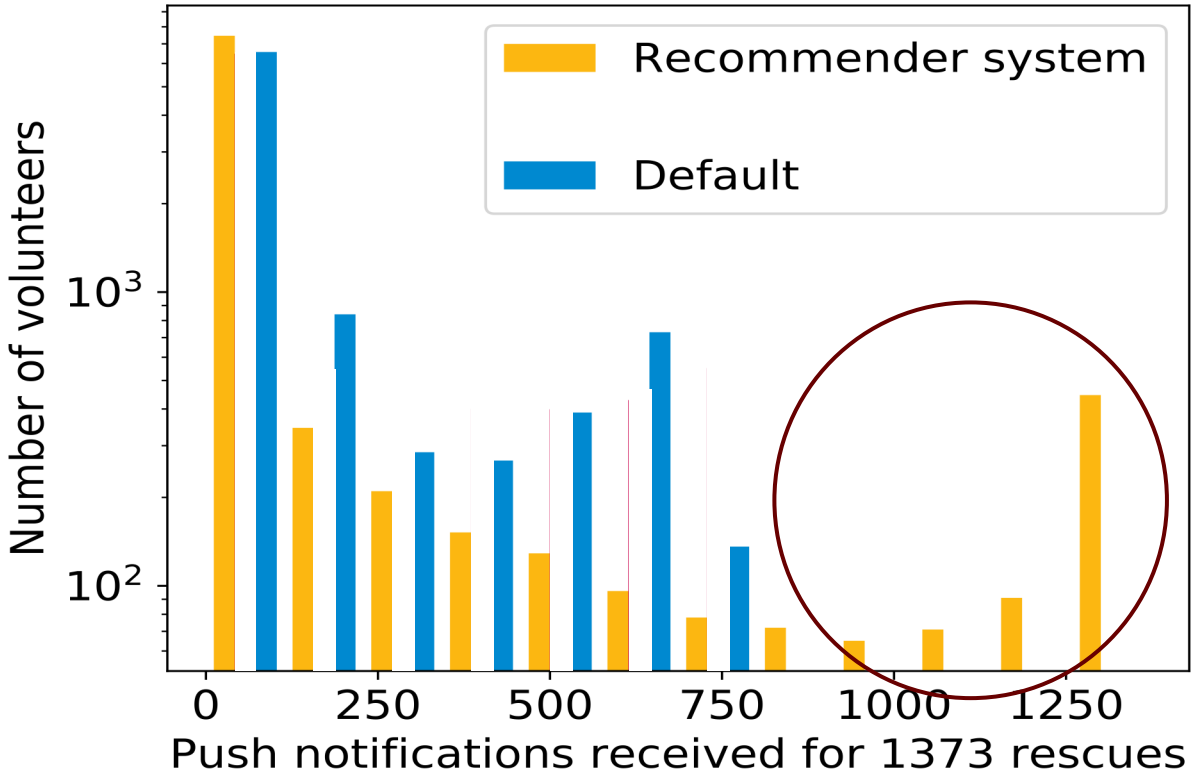
- Metric: Hit ratio at k (HR@k): % of rescues that are claimed by the k volunteers notified
- $k = 964$ to match the default INS

Model	HR@k (SD)
NN	**0.7269 (0.0310)**
RF(1:1)	0.5989 (0.0395)
RF(1:20)	0.6035 (0.0511)
GBDT(1:1)	0.6235 (0.0549)
GBDT(1:20)	0.5394 (0.0152)
SM(1:1)	0.4996 (0.0005)
SM(1:20)	0.5219 (0.0125)
Default	0.4392 (N/A)

A Recommender System for Crowdsourcing Food Rescue Platforms. Zheyuan Ryan Shi, Leah Lizarondo, Fei Fang In WWW-21

Caveat with Rescue-Specific Notification

- ML model discovers some frequent volunteers and sends them notifications almost all the time



A Recommender System for Crowdsourcing Food Rescue Platforms. Zheyuan Ryan Shi, Leah Lizarondo, Fei Fang In WWW-21

ML + Online Planning

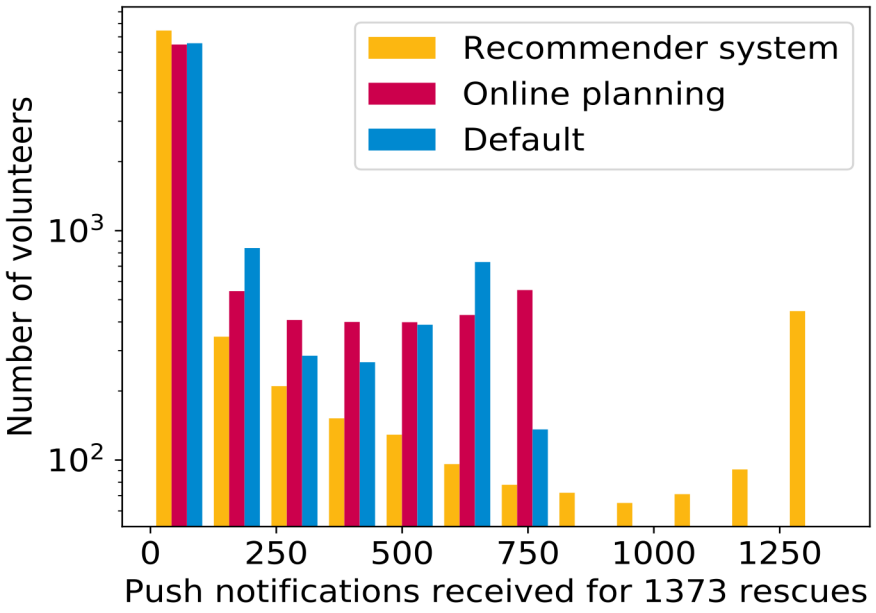
- Rather than greedily taking the top k volunteers, we enforce a constraint such that each volunteer receives at most L notifications per day

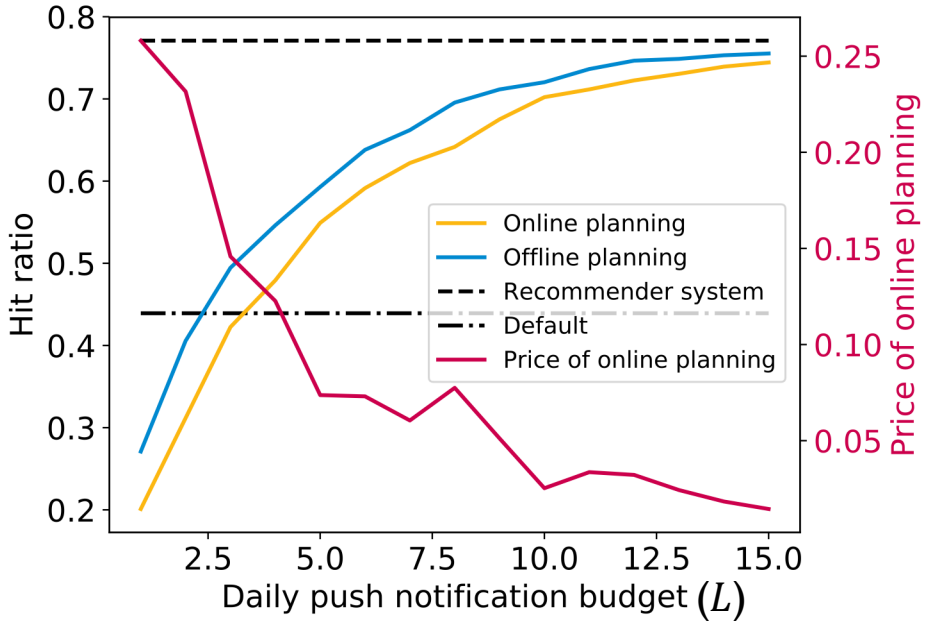
- For current rescue i, determine who to send notifications to by planning with an projected set of future rescues R

A Recommender System for Crowdsourcing Food Rescue Platforms. Zheyuan Ryan Shi, Leah Lizarondo, Fei Fang In WWW-21

Online Planning-Based Rescue-Specific Notification

- Avoid the over-concentration with $L = 5$
- HR@k = 0.645, much better than current practice

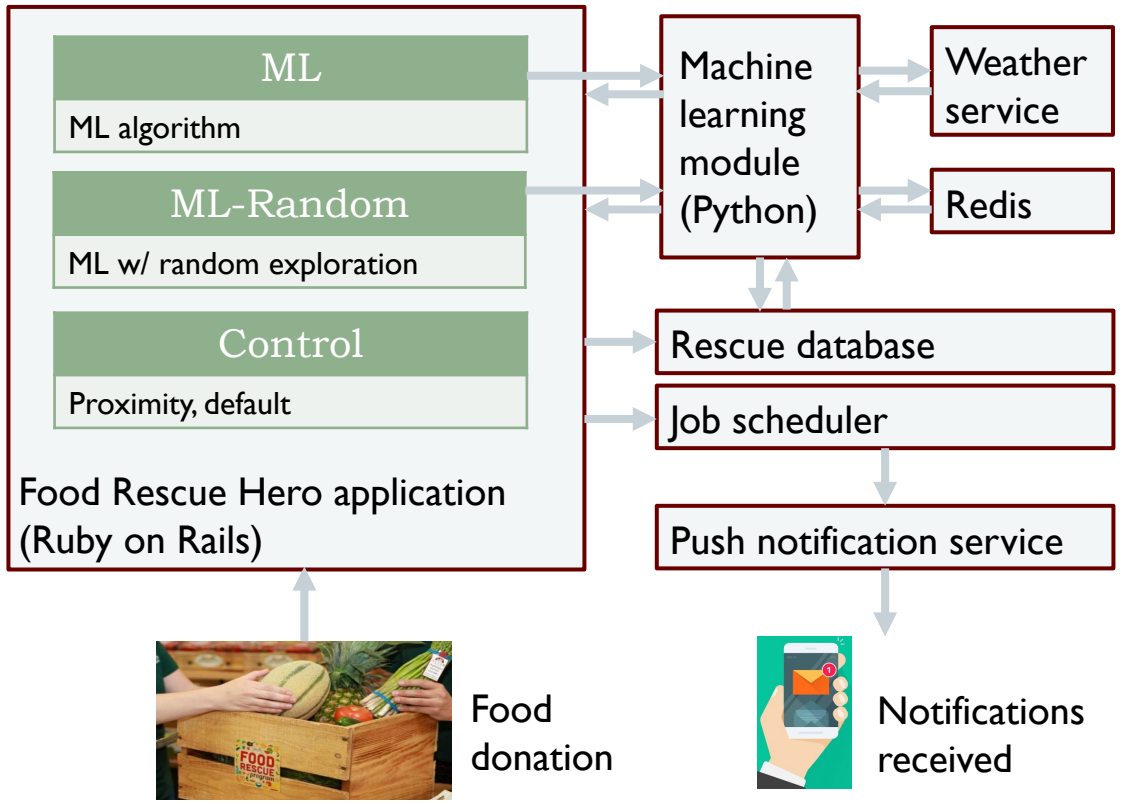




A Recommender System for Crowdsourcing Food Rescue Platforms. Zheyuan Ryan Shi, Leah Lizarondo, Fei Fang In WWW-21

Deployment is highly nontrivial!

- Time: May 2022 – June 2022
- Scope: All food donations in the Greater Pittsburgh area
- Environment: Real-time notification module in a Ruby on Rails application



First ever RCT on the platform!

ML model significantly improved the hit rate

Control
Proximity, default

ML
ML algorithm

ML-Random
ML w/ random exploration

Results

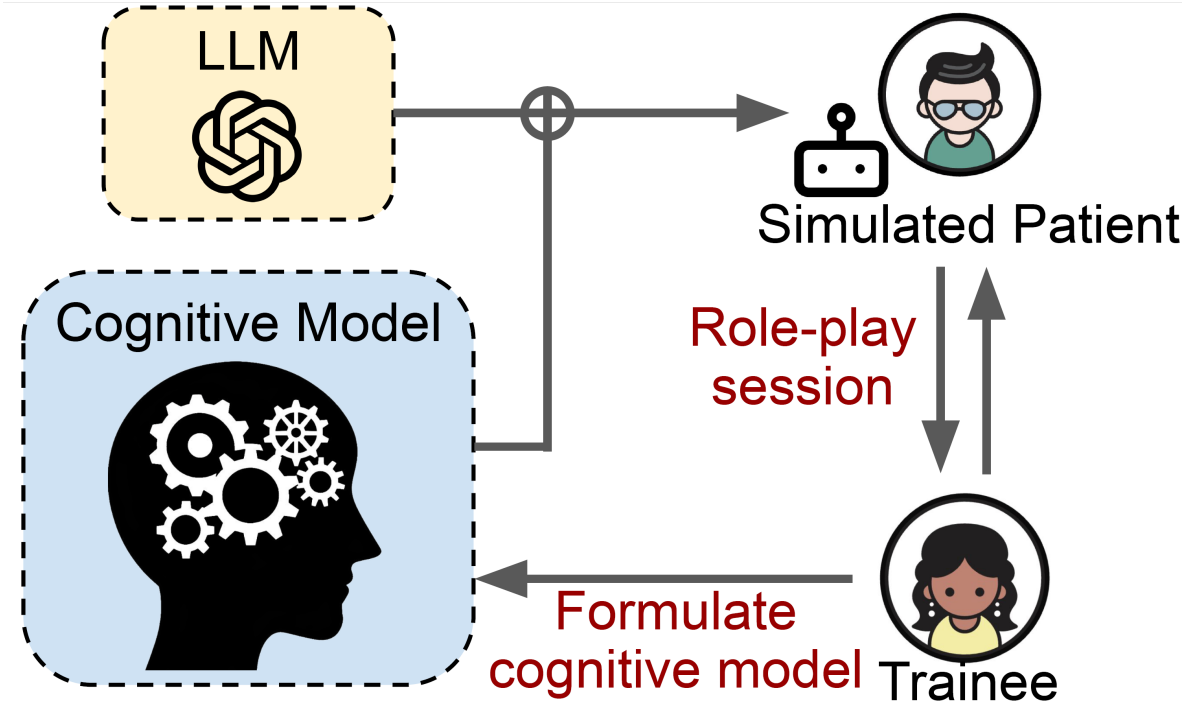
Condition	Hit Ratio (p-value w.r.t. control)	Claim Ratio (p-value w.r.t. control)
Control	0.468	0.807
ML	0.651 (0.001)	0.882 (0.047)
ML-Random	0.489 (0.696)	0.844 (0.317)

- The ML model also significantly improved the claim rate.

Today's Lecture

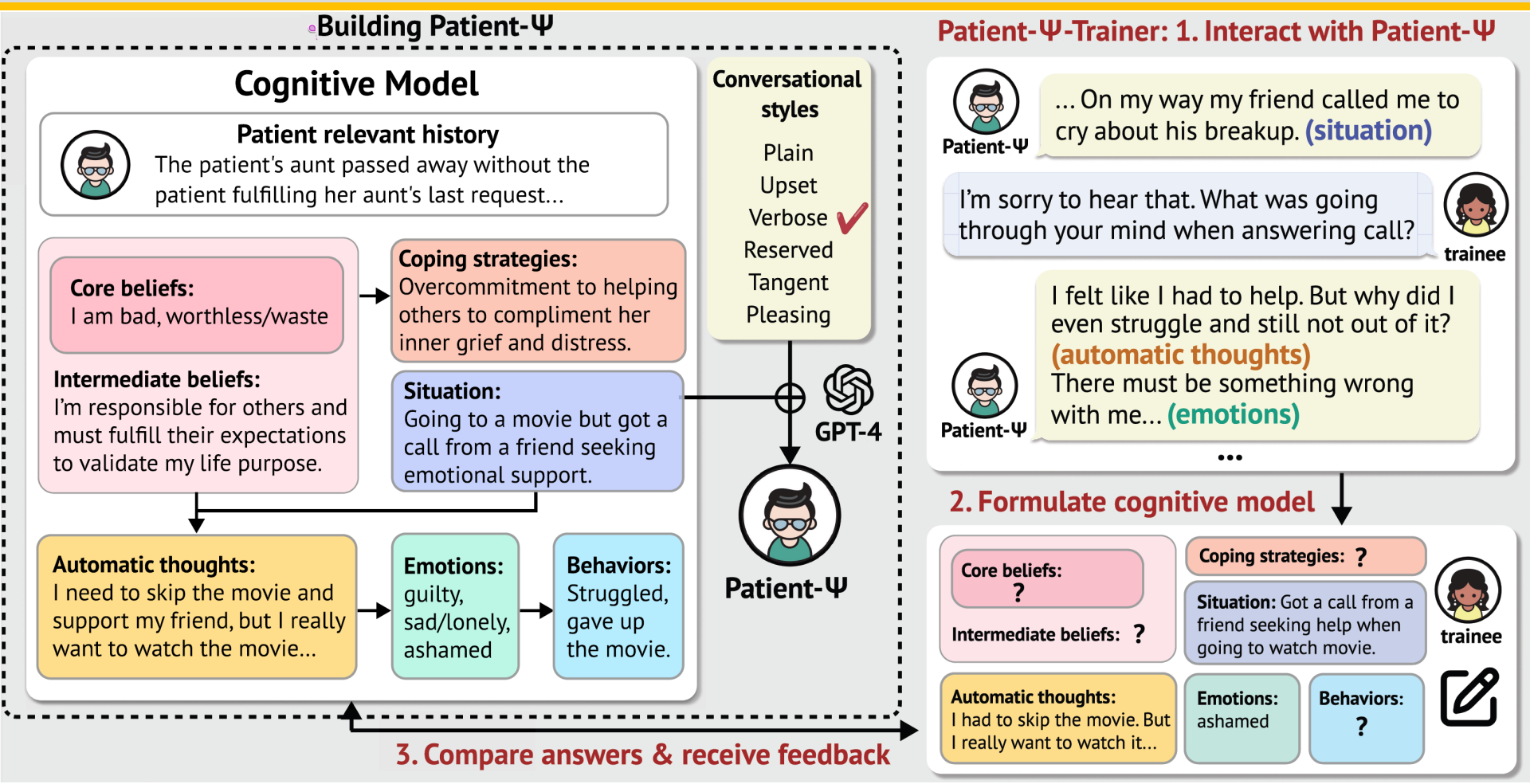
- Overview of AI for Social Good
- Example projects
  - PAWS and NewsPanda: AI for Environmental Conservation
  - Improving Operational Efficiency of Food Rescue Platforms
  - AI for Mental Health and Behavioral Health
- Summary

Simulated Patient for Cognitive Behavior Therapy (CBT) Training



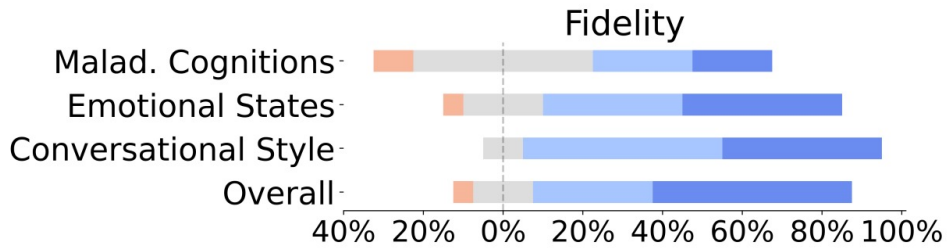
PATIENT-Ψ: Using Large Language Models to Simulate Patients for Training Mental Health Professionals. Ruiyi Wang, Stephanie Milani, Jamie C. Chiu, Jiayin Zhi, Shaun M. Eack, Travis Labrum, Samuel M. Murphy, Nev Jones, Kate Hardy, Hong Shen, Fei Fang, Zhiyu Zoey Chen. EMNLP 2024

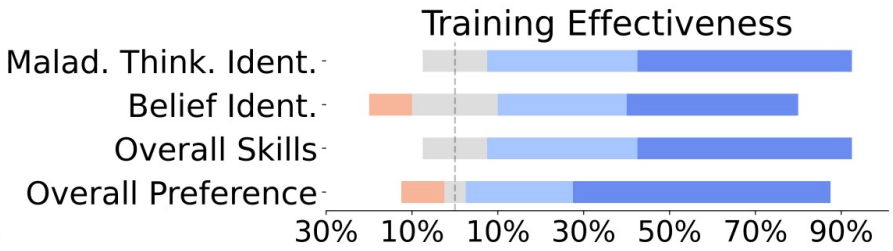
Simulated Patient for Cognitive Behavior Therapy (CBT) Training



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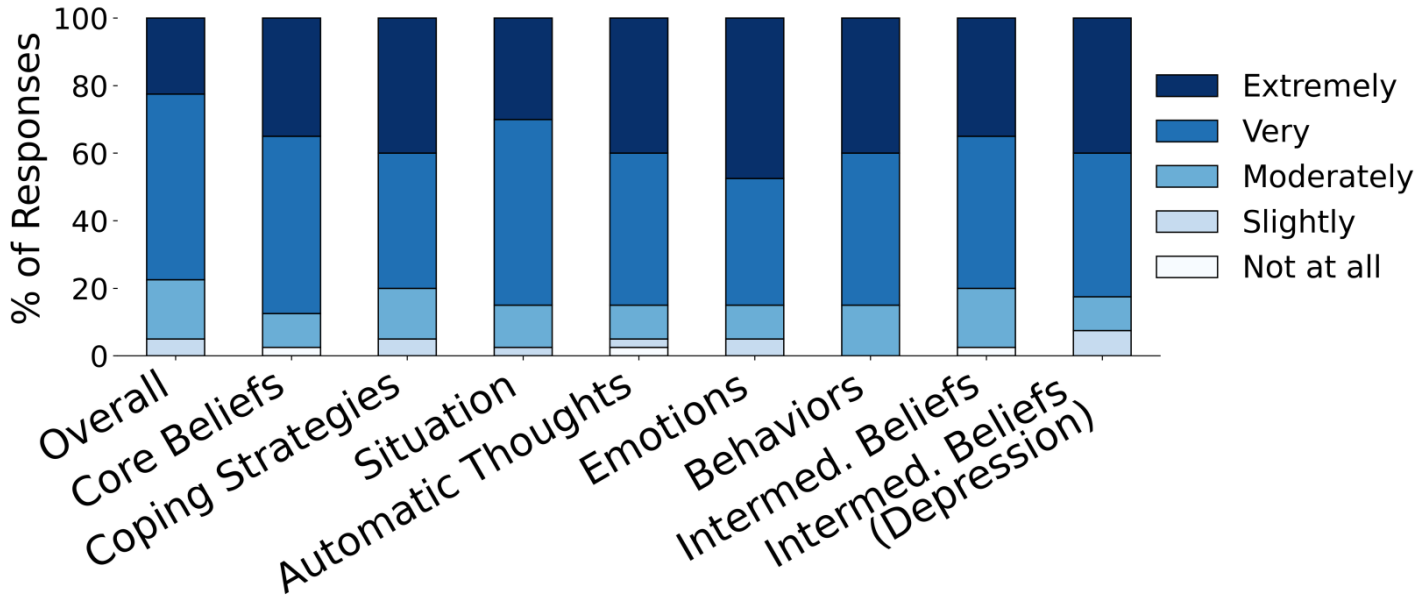
Simulated Patient for Cognitive Behavior Therapy (CBT) Training





GPT-4 is much better | GPT-4 is somewhat better | About the same | Patient-Ψ(-Trainer) is somewhat better | Patient-Ψ(-Trainer) is much better

Patient Ψ accurately in reflecting the reference cognitive model



Challenges in Peer-Run Behavioral Health Organizations

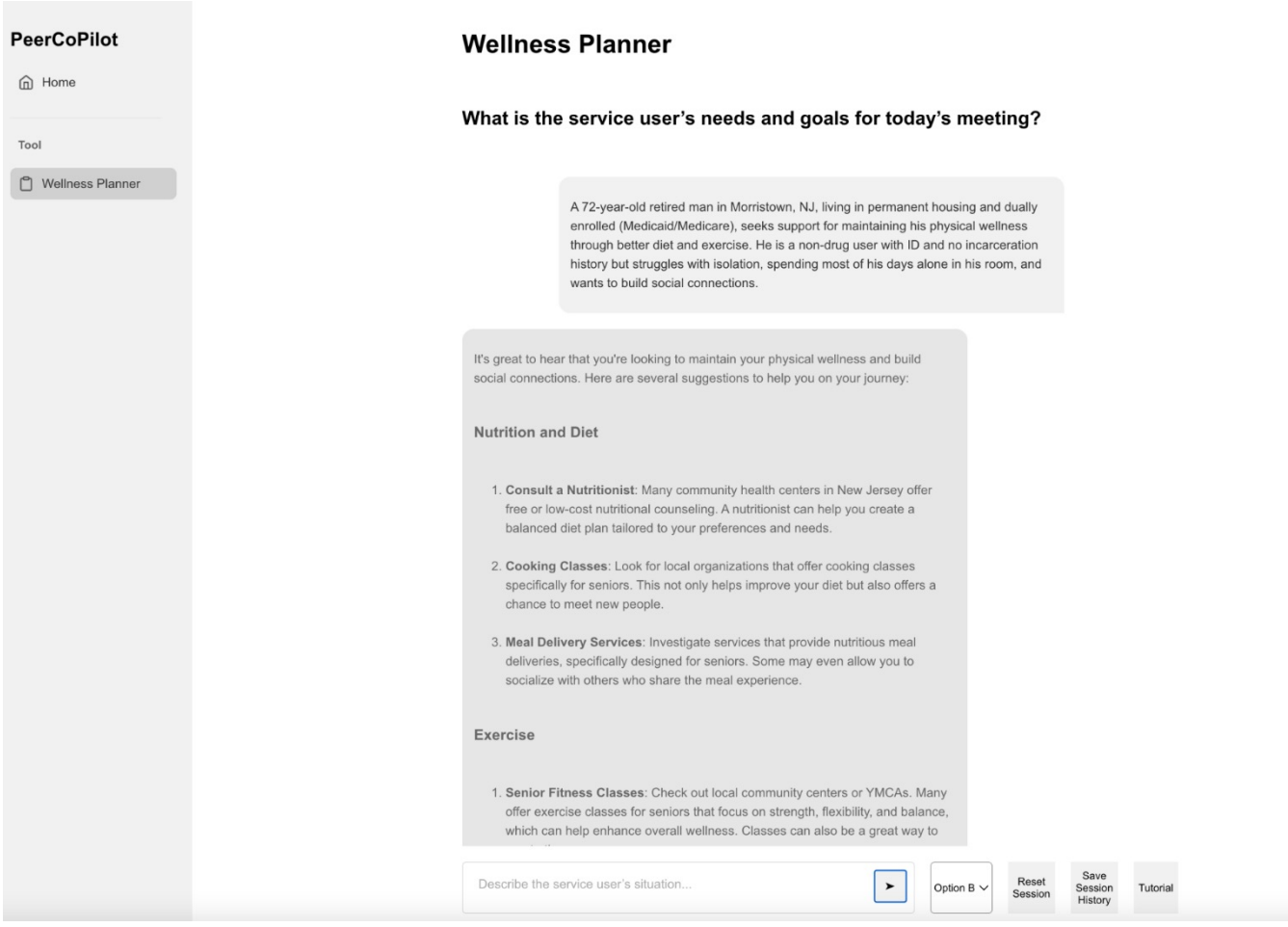
- Behavioral health conditions cost $80B+ annually
- Our collaborator, Collaborative Support Programs of New Jersey (CSPNJ), supports individuals facing behavioral health challenges, along with poverty, unemployment, and housing instability
- In 2024, CSPNJ served 675 service users in supportive housing, 700 in respite programs, and 15–100 daily at 18 wellness centers, but faced limited staff to meet growing, complex needs



Image Source: Collaborative Support Programs of New Jersey (CSPNJ)

PeerCoPilot

- PeerCoPilot: support CSPNJ's peer providers by enhancing efficiency, reducing cognitive burden, and improving service quality



PeerCoPilot

1 Service user situation



"A 19-year-old male in Newark, NJ, is living in temporary housing while working a construction job. He is without ID and seeks help with his housing situation and applying for SSI. While undiagnosed, his experiences/challenges are consistent with PTSD and he has a marked trauma history"

2 Extract & incorporate trusted resources

RAG-based Resource Recommendation

Benefit Eligibility Checker

Wellness Planner

Follow-up Questions

3 Wellness plan

☆ Goals

Find long-term housing and apply for identification, as it's needed to apply for SSI

Benefit analysis

You might be eligible for SSI, but it depends on how much income you make, both in work and non-work.

Resources

The Newark Housing Authority provides information on housing, including tips on applying for Section 8 housing...

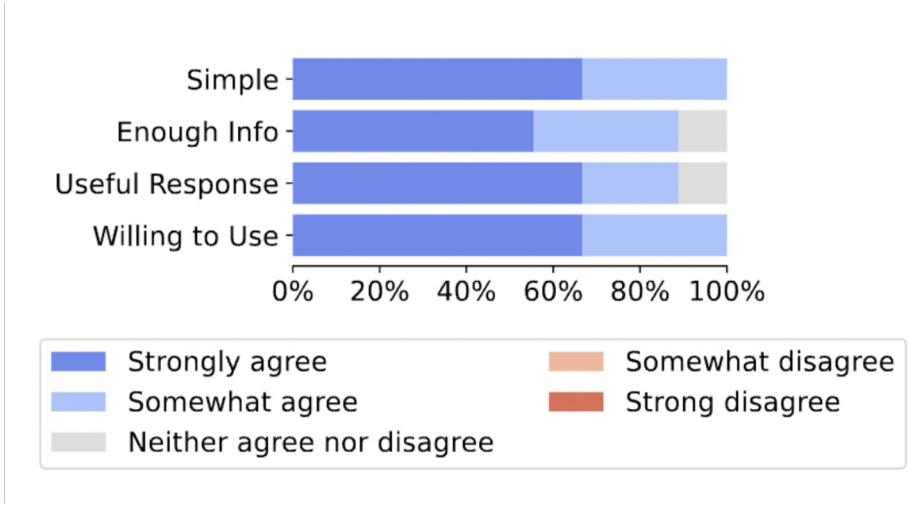
Questions

1. Do you have any non-work income?
2. Have you applied for SSI previously?

User Study and Human Evaluation

- 12 onsite sessions + 12 zoom sessions
  - 15 peer providers, 6 service users
- Key Findings
  - PeerCoPilot provides more reliable & specific information than baseline
  - All peer providers & service users willing to use PeerCoPilot





	Contact Provided	Bad Link
PeerCoPilot	100%	0%
Baseline	56%	11%

Challenges of Doing Research in AI for Social Good

- Non-research obstacles on the path to deployment
- Technical novelty vs. practical value
- Publication venue and review criteria

Lessons Learned

- 1. For work that requires field test or deployment, minimizing the need for extra equipment/effort
  - Start with current practice
  - Start with simple, assistive tool rather than building something that is completely new
  - Reduce the effort of collaborating partners that goes beyond their routine work

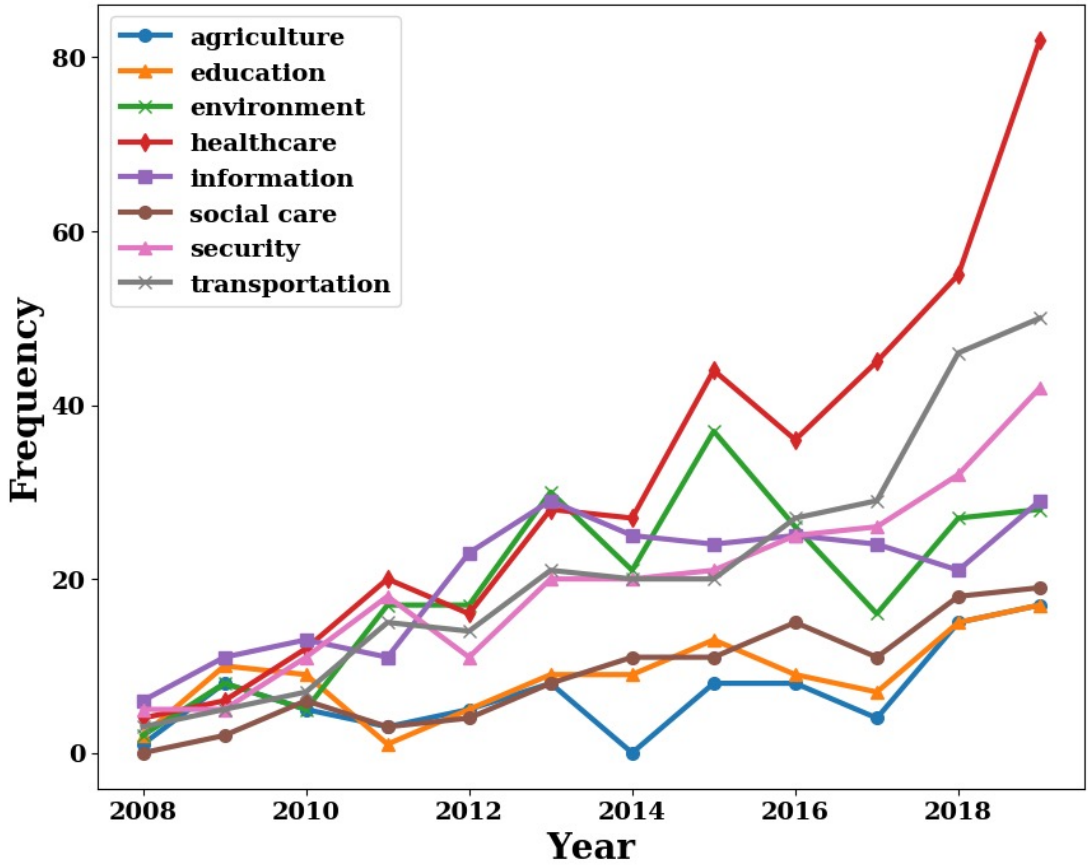
Lessons Learned

- 2. First-hand immersion in the domain/problem is critical to understanding the context and accelerating the development process
  - Also important in the field test/deployment stage to understand the limitations and make improvements

- 3. Visualizing the model / solution is important for communication and technology adaptation

Discussion: Propose an AI for Social Good project

- Pick an application domain, find an existing government agency/non-profit organization, and propose a potential AI solution that can be used by the agency/organization



Summary

- AI for Social Good

- Example projects
  - AI and Environmental Conservation, Food Rescue, Simulated Patient for CBT Training

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Acknowledgment

- Advisors, postdocs, students and all co-authors!

- Collaborators and partners











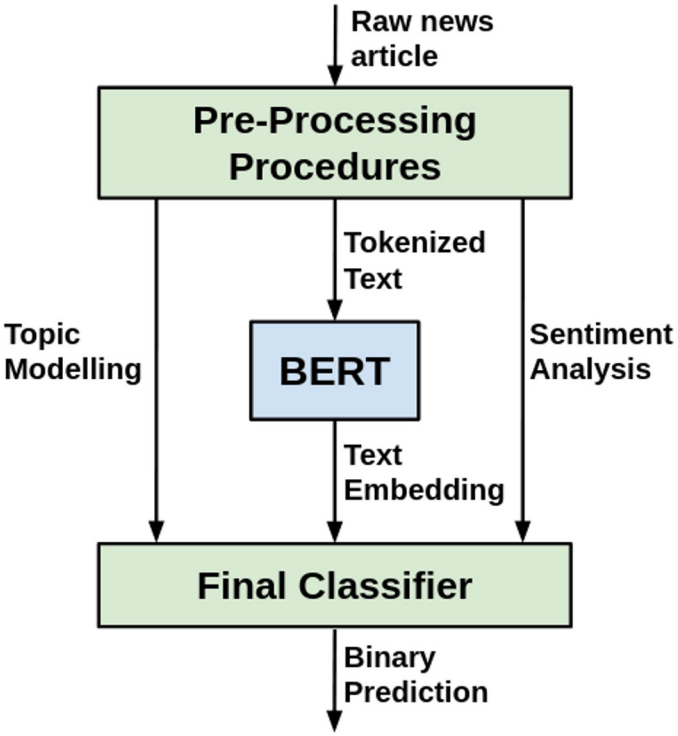


- Funding support



Backup Slides

Relevance Classification Module



We include the following features for each article:

- BERT embedding
- Sentiment analysis score
- Topic modelling vector

Prediction is done along two dimensions: **conservation relevance** and **infrastructure relevance**

NewsPanda: Relevance Classification Module

Model	Acc.	P	R	F1
Keyword	0.820 (n/a)	0.317 (n/a)	0.634 (n/a)	0.423 (n/a)
LSTM	0.711 (0.068)	0.495 (0.097)	0.511 (0.129)	0.504 (0.070)
GRU	0.729 (0.054)	0.422 (0.110)	0.505 (0.139)	0.475 (0.067)
BERT	0.860 (0.014)	0.708 (0.032)	0.704 (0.036)	0.706 (0.015)
RoBERTa	0.867 (0.009)	0.705 (0.044)	0.743 (0.041)	0.721 (0.025)
NEWSPANDA	**0.877** (0.013)	**0.729** (0.032)	**0.801** (0.051)	**0.744** (0.026)

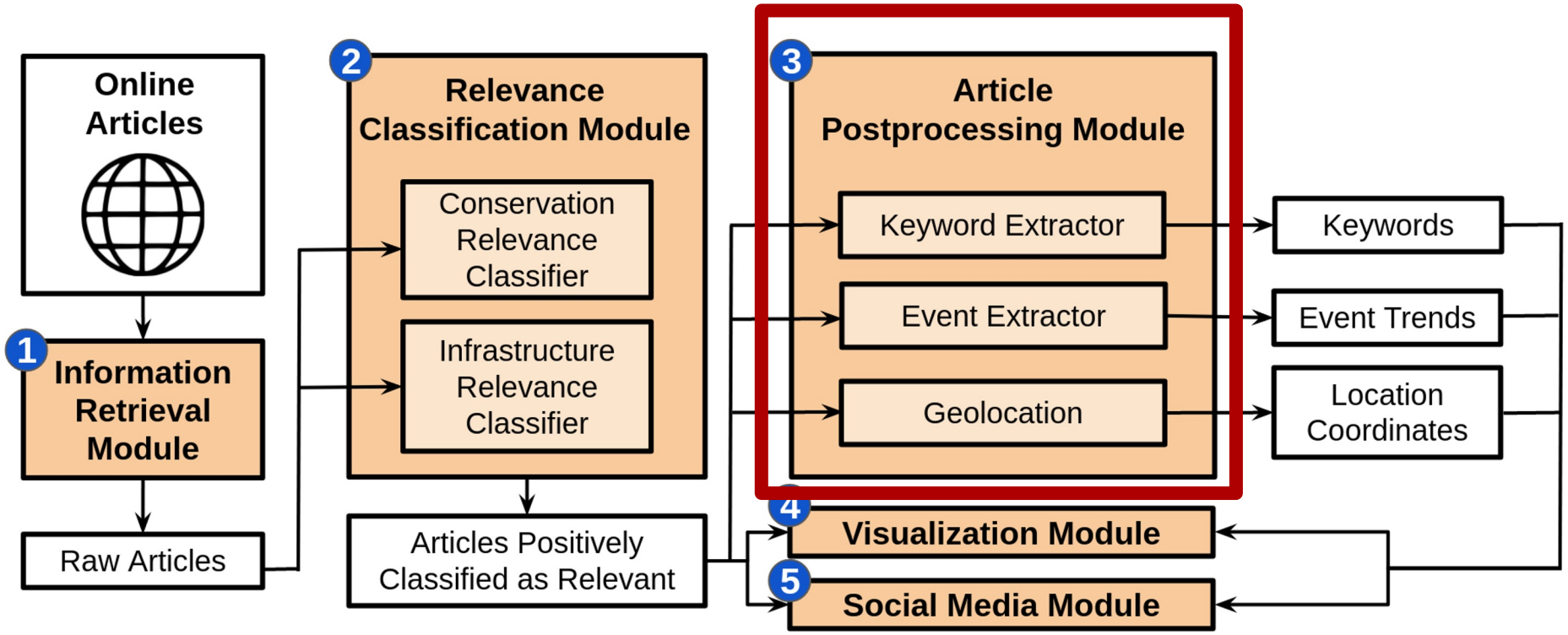
(a) Scores for *Conservation Relevance*

Model	Acc.	P	R	F1
Keyword	**0.947** (n/a)	0.250 (n/a)	0.455 (n/a)	0.323 (n/a)
LSTM	0.908 (0.027)	0.566 (0.160)	0.537 (0.088)	0.554 (0.065)
GRU	0.895 (0.022)	0.544 (0.109)	0.557 (0.123)	0.553 (0.109)
BERT	0.922 (0.018)	0.840 (0.154)	0.745 (0.152)	0.771 (0.096)
RoBERTa	0.916 (0.021)	0.794 (0.091)	0.809 (0.064)	0.799 (0.041)
NEWSPANDA	0.941 (0.018)	**0.880** (0.097)	**0.821** (0.051)	**0.850** (0.043)

(b) Scores for *Infrastructure Relevance*

NewsPanda performs the best across all the models and baselines.

NewsPanda pipeline



Keyword extraction

(example paragraph)

A 45-year-old man from Chamrajanagar in Karnataka was arrested by the Forest Department for attempting to hunt wild animals and also setting fire in forest areas in the Talavadi Forest Range in the Sathyamangalam Tiger Reserve here.



Species- **tiger**
Terrestrial Habitat - **forest**

	A	B
1	Atmosphere	carbon
2	Atmosphere	CH4
3	Atmosphere	CO2
4	Atmosphere	methane
5	Atmosphere	nitrogen
6	Conservation Keyword	adaptation
7	Conservation Keyword	adaptive management
8	Conservation Keyword	alternative livelihoods
9	Conservation Keyword	animal identification
10	Conservation Keyword	animal traceability
11	Conservation Keyword	animal welfare
12	Conservation Keyword	benefit sharing

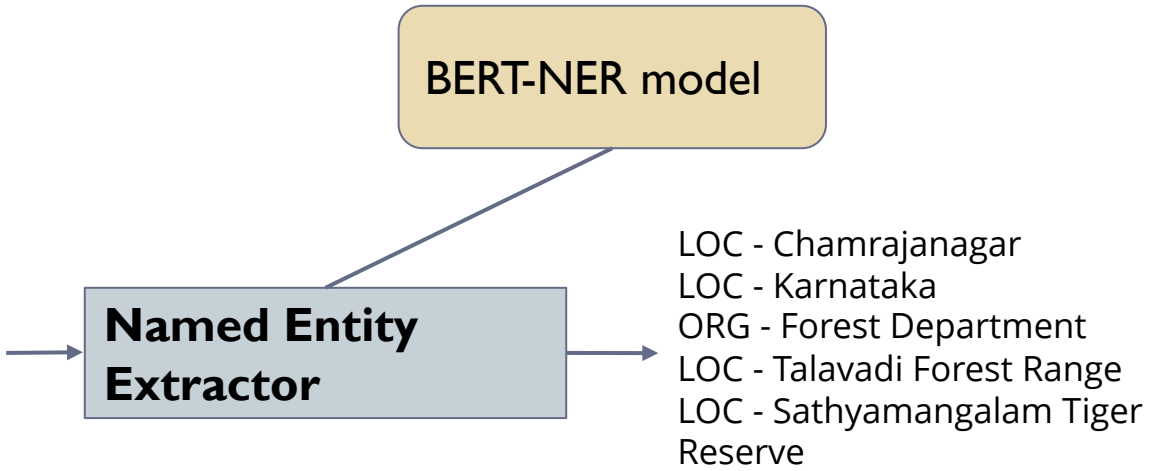
We use a list of around 1000 keywords (level 1 and level 2), then check for matches in the text.

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Named entity recognition

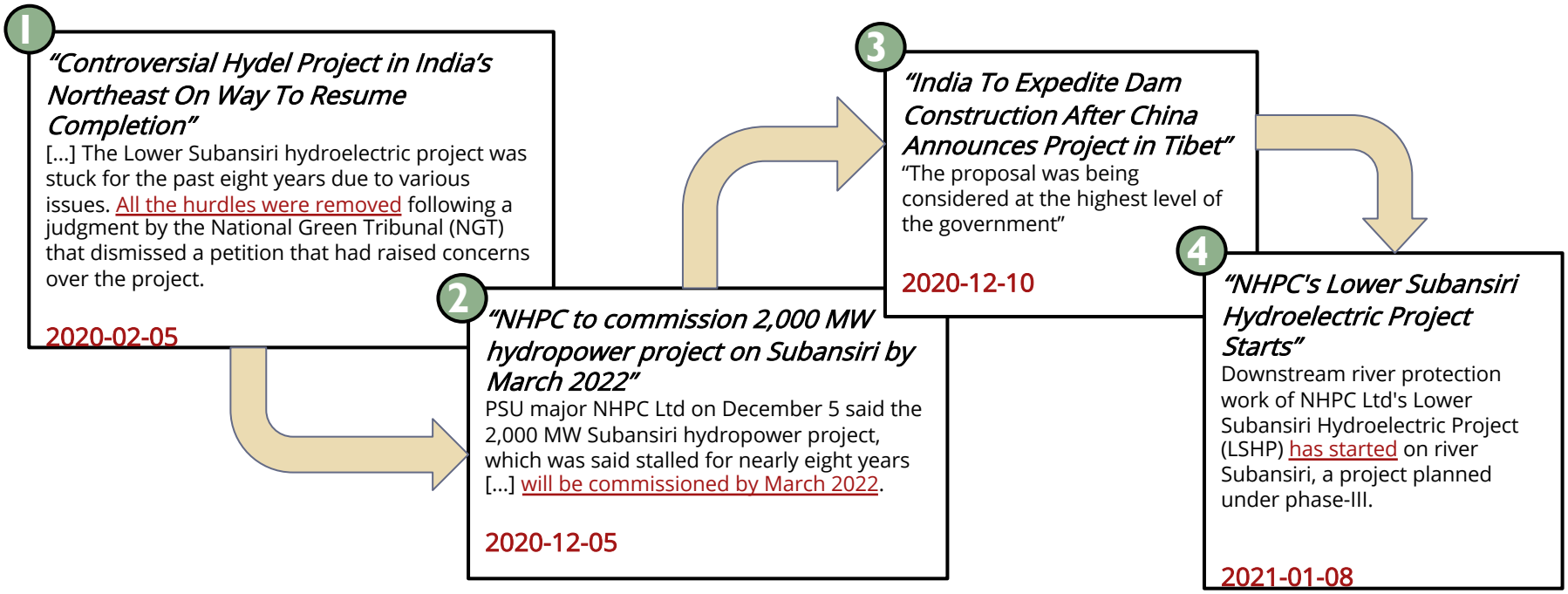
(example paragraph)

A 45-year-old man from Chamrajanagar in Karnataka was arrested by the Forest Department for attempting to hunt wild animals and also setting fire in forest areas in the Talavadi Forest Range in the Sathyamangalam Tiger Reserve here.



Event timeline

We search for entity **"Subansiri"**, then filter with the keyword "**hydroelectric**".



Geolocation

- Important to integrate into WWF's GIS systems
- Use a directory of conservation sites from WWF to map articles to their coordinates
- If there is no match in directory, we use the geopy package

Deployment

- NewsPanda has been deployed by WWF teams in India, Nepal, and the UK since February 2022

	Conservation			Infrastructure		
	P	R	F1	P	R	F1
WWF India	0.849	0.605	0.706	0.462	0.250	0.324
WWF Nepal	0.895	0.917	0.906	0.923	0.308	0.462
WWF UK	0.879	0.823	0.850	1.000	0.455	0.625

High precision ☺

Low recall for infrastructure: misses out on potential articles ☹

- Positive feedback from WWF after several months of deployment

> "You're giving us a bunch of articles... over 50 articles a week. We had two interns who spend 2-3 days a week on this and would only give us seven to ten articles. So there is a huge bump in efficiency right there in itself."

> "The data that you're sharing give a global perspective. It is very useful to understand the upcoming projects or mitigation measures that are being adopted on a global scale. So it helps us be informed."

NewsSerow: Leverage LLMs for Low-Resource Language

- With only a few examples, NewsSerow achieves comparable performance to fine-tuned models (which uses much more training data) for Nepali articles

Model	Nepal/Nepali		
	Precision	Recall	F1-Score
Zero-shot Models			
GPT-3.5-Turbo	0.79 (0.02)	0.32 (0.01)	0.46 (0.01)
Few-shot Models			
mBERT	0.26 (0.03)	0.70 (0.12)	0.37 (0.01)
XLM-R	0.35 (0.04)	0.60 (0.23)	0.43 (0.06)
GPT-3.5-Turbo	0.68 (0.05)	0.58 (0.07)	0.62 (0.02)
NewsSerow	0.88 (0.03)	0.58 (0.01)	**0.70** (0.01)
Fine-tuned Models			
mBERT	0.77 (0.06)	0.57 (0.05)	0.65 (0.02)
XLM-R	0.70 (0.05)	0.71 (0.04)	**0.70** (0.01)
Translation Test	0.74 (0.16)	0.64 (0.15)	0.66 (0.06)

NewsPanda++ Dashboard for Nepal

Dashboard All Articles View Logout

Dashboard

Landscape Overview



222
ARTICLES

54
CONSERVATION EVENT RELATED ARTICLES

Chitwan Annapurna Landscape
CRITICAL LANDSCAPE

15
INFRASTRUCTURE EVENT RELATED ARTICLES

Other articles: 85

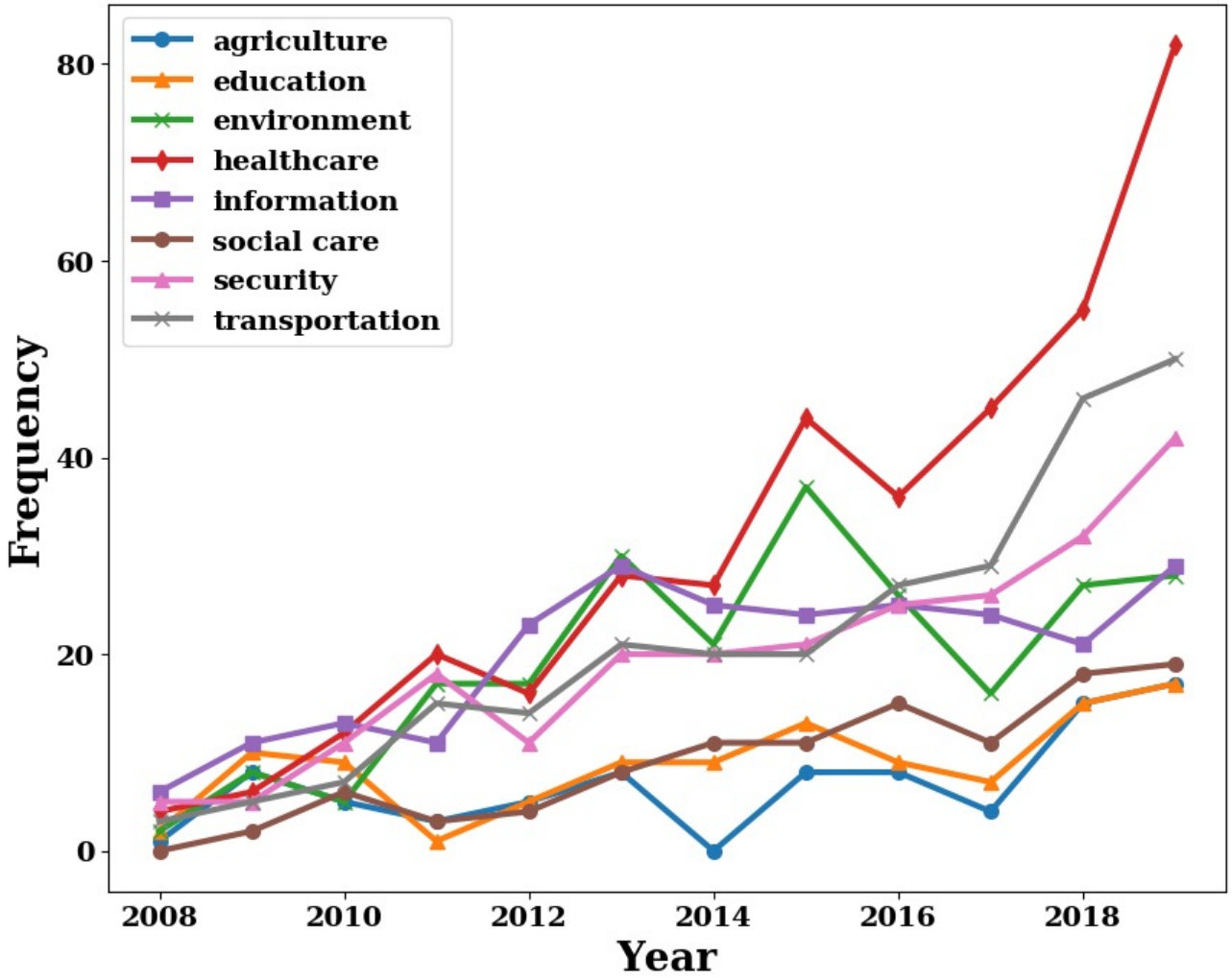
NewsPanda++ Dashboard for Nepal

Dashboard All Articles View All Events View Logout

All Events

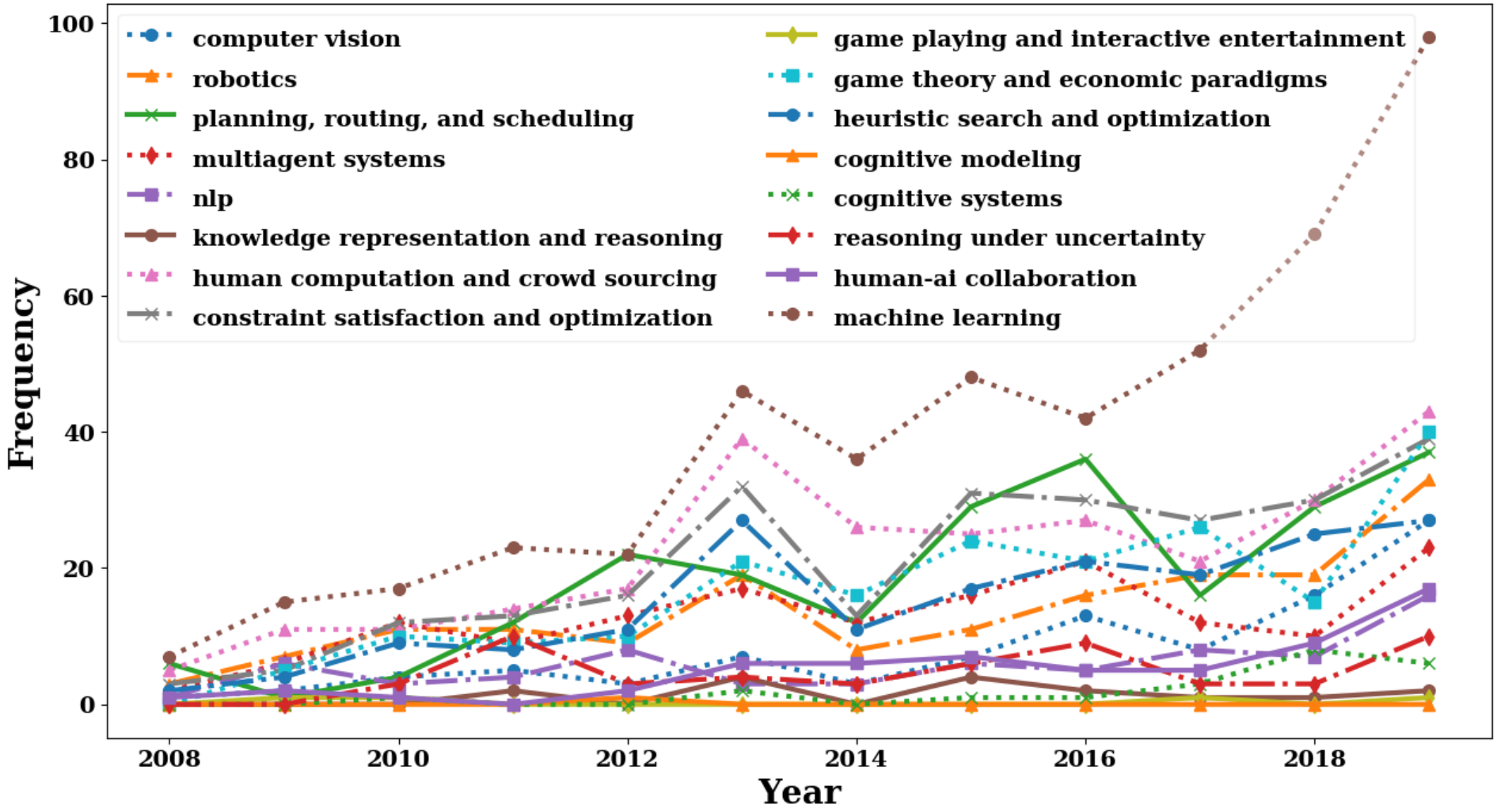
Event ID	Title	Sources	Category	Summary	Recommended Actions
1	Proposal to create a new tiger reserve in Suhelwa Wildlife Sanctuary	https://twitter.com/MadhuriDaksha/status/1804895658987712779 https://twitter.com/theupindex/status/1804883494482661488	Habitat Conservation	Balrampur, UP, is poised to become home to the state's fifth tiger reserve, following a recent census confirming tiger presence in Suhelwa Wildlife Sanctuary, located along the Indo-Nepal border.	1. Support local wildlife conservation initiatives. 2. Participate in awareness programs about tiger conservation.
2	Nepal's wildlife threatened by new hydropower and hotel developments	https://twitter.com/dailynewsnp/status/1803444427194769742 https://twitter.com/Planet_Rhino/status/1792277796439699705	Infrastructure Events	Concerns are being raised over planned hydropower dams and new regulations allowing hydropower and hotel projects in Nepal's sacred rivers and protected nature reserves, potentially endangering indigenous communities, wildlife habitats, and several endangered species.	1. "Raise awareness about the impact of hydropower dams on Nepals environment." 2. Petition against the regulations permitting projects in protected areas.
3	Wildfires severely impact Nepal's forests and wildlife	https://twitter.com/firehorse249791/status/1787406857234972747 https://twitter.com/mjshree342/status/1786753147379011864 https://twitter.com/firehorse249791/status/1785354305920229529	Infrastructure Events	Nepal is currently facing severe wildfires that have destroyed 5,000 hectares of forest in Banke and threatened homes, wildlife, and people's lives across the region. These fires have also narrowly missed a flagship hospital near Kathmandu.	1. Raising awareness about the Nepal wildfires on social platforms 2. Donating to organizations helping in wildfire relief and rehabilitation in Nepal
4	Bilateral Meeting on Curbing Illegal Wildlife Trade Between India and Nepal	https://twitter.com/bsi_moefcc/status/1805978606943850805 https://twitter.com/bsi_moefcc/status/1805978602648776906 https://twitter.com/SAWENsec/status/1805855689819971990 https://twitter.com/TRAFFIC_India/status/1805852717677691264	Illegal Wildlife Trade	A bilateral meeting between India and Nepal was held in Siliguri, India to discuss the challenges and solutions related to illegal wildlife trade and promote stronger collaboration between the two countries. More than 30 officials attended the meeting, which featured talks on the role of scientific authorities in CITES implementation and the facing issues.	1. Promote cross-border collaborations for wildlife conservation 2. Strengthen the implementation of CITES

Application Domains



Artificial Intelligence for Social Good: A Survey Zheyuan Ryan Shi, Claire Wang, Fei Fang

AI Techniques



Artificial Intelligence for Social Good: A Survey Zheyuan Ryan Shi, Claire Wang, Fei Fang

Application Domains & AI Techniques

